BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

033/1 **BIOLOGY 1**
(For Both School and Private Candidates)

Time: 3Hours **Wednesday, 26th June 2019 a.m.**

Instructions

1. This paper consists of section A, B, and C.
2. Answer **all** questions in sections A and B **one (1)** question form section C.
3. Except for diagram that must be drawn in pencil, all writing should be in blue or black ink.
4. Calculators and cellular phones are not allowed in the **Examination room.**
5. Write your **Examination Number** on every page of your answer booklet(s).

SECTION A (20 Marks)

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet provided.
   (i) The rate of photosynthesis is greatly influence by the following factors except.
      A. Temperature.
      B. Water supply.
      C. Soil fertility.
      D. Carbon dioxide.
      E. Light intensity.
   (ii) A strong tissue that holds muscles to bones at joints is called.
      A. Ligament.
      B. Cartilage.
      C. Connective tissue.
      D. Tendon.
      E. Synovial membrane.
   (iii) The absence of one of the following elements makes a plant fail to produce chlorophyll.
      A. Magnesium.
      B. Sulphur.
      C. Chlorine.
      D. Calcium.
      E. Iron.
   (iv) The part of a brain responsible for reasoning and imagination is known as.
      A. Spinal cord.
      B. Medulla oblongata.
      C. Cerebrum.
      D. Cerebellum.
      E. Brain chamber.
   (v) The last stage of Mitosis during growth of an organism is.
      A. Metaphase.
      B. .Telophase
      C. Transpiration.
      D. Prophase.
      E. cytokinesis.
   (vi) By which process does carbon dioxide pass from the blood to alveoli of the lungs?
      A. Osmosis.
      B. Oxidation.
      C. Translocation.
      D. Diffusion.

E. Respiration.

(vii) The following sequence of organisms represents.

Earthworm ⟶ Tilapia ⟶ Human being ⟶ Bacteria

A. Food web.
B. Food chain.
C. Ecosystem.
D. Community.
E. A pyramid.

(viii) Which of the following organism belong to phylum protozoa?

A. Mushroom.
B. Spirogyra.
C. Amoeba.
D. Funaria.
E. Tick.

(ix) In which category the malpighian layer of mammalian skin is placed?

A. Tissue.
B. Organ.
C. Cell.
D. System.
E. Organism.

(x) A group of animals that maintain the body temperature at more or less constant is-

A. Regulators.
B. Cold blood animals.
C. Warm blood animals.
D. Poikilotherms.
E. Homoeotherms.

2. Match the phrases in **List A** with the responses in **List B** by writing the letter of the correct response from **List B** beside the item number of **List A** in your answer sheet provided.

List A	**List B**
i. A hard outer cover of animals such as crustaceans and arachnids.	A. Flagella movement.
ii. Found at the elbow and knee and work like a hinge if door to allow movement in one plane.	B. Ciliary movement.
	C. Tendon.
	D. Ligament.

iii. Found in the heart and are special in that they contract and relax without getting fatigue. iv. Movement of plant towards or away from an environment stimulus. v. Type of movement exhibited by the organisms such as euglena. vi. Rigid structure made of calcium deposits and living cells. vii. Is a location where bones meet. viii. A strong bundle of fibres which attaches muscle to bone. ix. Bundle of fibres around a joint in the skeleton which hold the bone in place and help to prevent dislocation. x. Bundles of fibrous tissue in animals which has the ability to contract thereby producing movement.	E. Honge joint. F. Exoskeleton. G. Cardiac muscle. H. Skeletal muscle. I. Tropical movement. J. Nastic movement. K. Bones. L. Joints. M. Endoskeleton. N. Ball and socket. O. Joint. P. Hydrotropism. Q. Skeletal movement. R. Muscle.

SECTION B (60 MARKS)

Answer **all** questions in this section.

3. (a) State four (4) Precautions for proper use of a Microscope.
   (b) Differentiate a test experiment from a control experiment.

4. (a) (i) What is gaseous exchange?
   (ii) Arrange the following in the correct order:
   Bronchi, pharynx, bronchioles, nasal passages, alveoli, trachea.
   (b) Name the structure concerned with gaseous exchange in
   i. A mammal
   ii. A young tadpole.
   iii. An insects.

iv. An earth worm
v. Spider.

5. (a) (i) What is a reflex action?
(ii) Give four (4) examples of reflex actions.
(iii) Explain the importance of reflex action to animals.

(b) (i) State two (2) differences between Nervous control and hormonal control.

(ii) Why a person manages to climb a big tree when suddenly see a big lion.

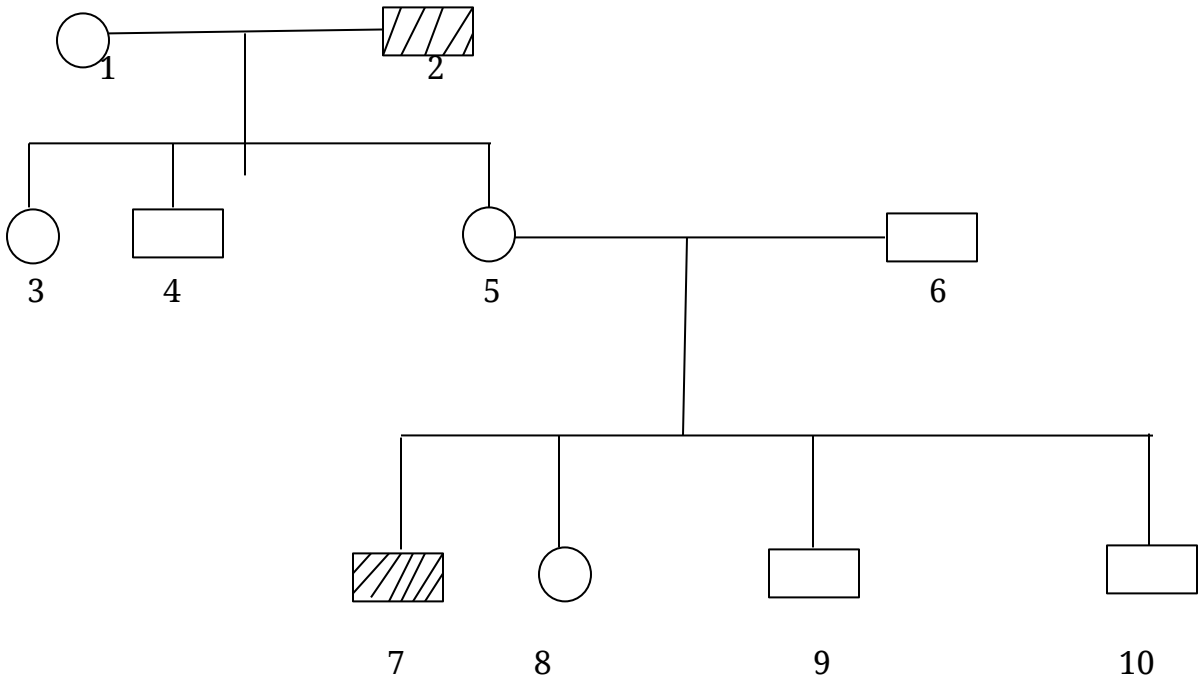
6. (a) Explain the two 2) types of mutation in living organisms.
(b) Give four (4) examples of genetic disorders caused by gene mutation.

7. (a) Explain the meaning of the following terms.
i. Transportation of Materials.
ii. Mass flow.
(b) Explain what would happen when a plant cell is placed in hypertonic solution.

8. Define the following terms as used in reproduction.
i. Fertilization.
ii. Ovulation.
(b) Name and explain how the two (2) types of twins occurs.

9. Study the following list of substances:
Water, Urea, Glucose, Calcium, Potassium and Sodium.
(a) Which of the substance will be excreted by a Mammal in a greater amount under each of the following?
i. Low temperature of the environment.
ii. Insufficient secretion of insulin.
iii. Consumption of large amount of proteins.
(b). Explain why one does not urinate frequently on a hot day.

10. (a) Define the following.
i. Genotype.
ii. Variation.
iii. Phenotype.
(b) Study the pedigree below and then answer the questions that follow.



KEY

○ = Females ▭ = Males

Open figure indicate normal phenotype; black figure indicate haemophilia individuals.

i. What is the probable genotype of 1?

ii. Is the answer to (i) above 100% certain? Explain.

iii. What is the genotype of 5 and 9?

iv. Justify your answer in (iii) above

v. If 3 marries normal man, what are the chances that will have a haemophilic son? Illustrate your answer.

SECTION C (20 Marks)

Answer **one (1)** question from this section.

11. Elaborate any three (3) traditional and three (3) modern methods used in food processing, preservation and storage.

12. Explain six (6) factors that affect the body immunity in human being.

13. Explain the functions of the major components of the human skeleton and their adaptations.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

033/2A **BIOLOGY 2A**

(ACTUAL PRACTICAL)
24 HOURS ADVANCE INSTRUCTIONS
(For Both School and Private Candidates)

ESSENTIAL:

01. Massive care must be taken not to give away these instructions to both candidates

and an authorized persons either directly or indirectly.

02. Please do not open the envelope containing the question papers before time of commencement of the Examination.

03. PREPARATIONS AND LABELLING OF SPECIMENS AND SOLUTIONS.

3.1 Preparation of solution / chemicals/ reagents.

(a) Prepare fresh solution of the Iodine solution to each candidates.

(b) Provide alcohol (ethanol) to each candidates.

(c) Detached a variegated leaf from potted plant and provided to each candidate.

3.2 LABELLING OF THE SPECIMEN

(a) Bread mould ---------- R

(b) Hibiscus flower ---------- S

(c) Bee ---------- T

(d) Millipede ---------- U

N.B: Provide all apparatuses indicated in 1st advanced instructions to each candidate.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

033/2A BIOLOGY 2A

(ACTUAL PRACTICAL)
(For Both School and Private Candidates)

Time: 2:30 Hours **Friday, 28th June 2019 a.m.**

Instructions

1. This paper consists of **two (2)** questions. Answer all questions.
2. Each question **carries** 25 marks.
3. Except for diagrams which must be drawn in pencil, **all** writings should be in blue or black ink.
4. Calculators and cellular phones are **not** allowed in the Examination Room.
5. Write your **Examination Number** on every page of your answer booklets.

1. You have been provided with variegated leaf and Iodine solution. Carefully follow the instructions and answer the question that follows.
   (i) Heat some water to boiling point in a beaker and then turn off the source of heat.
   (ii) Use forceps to dip the leaf in the hot water for about 30 seconds.
   (iii) Remove the leaf from the beaker.
   (iv) Push the beaker into the bottom of test tube and cover it with alcohol

(ethanol).

(v) Place the tube in hot water until alcohol boils with the leaf.

(vi) Remove the leaf from the tube containing ethanol and dip it into hot water.

(vii) Spread the decolorized leaf on the white tile and drop the Iodine solution on it.

(a) What is the aim of this experiment?

(b) Why leaf was dipped in hot water for 30 seconds?

(c) (i) Give reasons, why leaf was boiled in ethanol?

(ii) Why leaf was dipped once again in hot water?

(d) Give the interpretation of the results observed when few drops of Iodine solution were poured onto a decolorized leaf.

(e) What conclusion can be made from the result obtained in the above experiment?

(f) Name and define the biological process concerned with the above experiment.

2. You have been provided with specimens R, S, T and U.

(a) Study specimen R, S, T and U carefully then:

(i) Identify specimen R, S, T and U using their common names.

(ii) Draw a well labeled diagram of S.

(iii) Classify specimen S, T and U up to class level.

(iv) Differentiate between classes in which specimen T and U belong.

(b) (i) State the economic importance of specimen T.

(ii) Outline four (4) distinctive features of specimen T.

(iii) Name the kingdom and division in which the specimen R belongs.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT
FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

033/2A **BIOLOGY 2A**

(ACTUAL PRACTICAL)
(For Both School and Private Candidates)

Time: 2:30 Hours **Friday, 28th June 2019 a.m.**

Instructions

1. This paper consists of **two (2)** questions. Answer all questions.
2. Each question **carries** 25 marks.
3. Except for diagrams which must be drawn in pencil, **all** writings should be in blue or black ink.
4. Calculators and cellular phones are **not** allowed in the Examination Room.
5. Write your **Examination Number** on every page of your answer booklets.

1. You have been provided with solution.
   (a) Identify the food substances present in solution H by using the reagents provided. Tabulate your work for identifying food present
   (b) For each food substances identified in 1(a)

(viii) Name three (3) common sources
(ix) State their role in the body of human being
(x) For the food have you identified write their chemical formula

(c) On your identification of food present in sample H have been used the different chemicals and reagent explain briefly the following below
(i) When Hcl is added on your food sample they must be boiled
(ii) Benedict solution cannot be added in acidic media

(d)(i) Mention two enzymes which results same product in the digestive system of food you have identified.
(ii) Mention two adaptation for food digested in the mouth to occur and what type of digestion.

2. You have been provided with specimens A, N, H and Z.
(c) Use the hand lens to observe these specimens then
(v) Identify specimen A, N, H and Z by their common names.
(vi) Classify specimen A, N, H and Z to class level.

(d) (i) Mention three types of specimen A and their disease.
(ii) State habitats of specimen A

(c) Draw a neat, large and well labeled diagram of specimen H

(d) Construct simple food web by using the specimen have been provided on your table.

(e) The feeding relationship above become apex from bottom to the top, mention two reasons.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

062 **BOOK KEEPING**

(For Both School and Private Candidates)

Time: 3 Hours **Monday, 01st July 2019 a.m.**

Instructions.

1. This paper consists of section A, B and C with a total of **seven (7)** questions.
2. Answer **all** questions.
3. Calculator, cellular phones and any unauthorized material are **not** allowed in the examination room.
4. Write your **Examination Number** on every page of your answer booklets/sheet(s).

SECTION A (20 MARKS)

Answer **all** questions in this section

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet/sheet(s) provided.

   i. Given the cost of goods sold is shs.800, 000/= and margin of 20%, what is the percentage of mark up?
      A. 40 B. 25
      C. 50 D. 75

   ii. The manufacturing account seeks to arrive at.
      A. Production costs paid in the year.
      B. Cost of goods produced.
      C. Cost of goods sold.
      D. Gross profits of goods sold.
      E. Net profit of goods produced.

   iii. Where there is no partnership agreement, profits and losses must be shared.
      A. In the same proportion as capital.
      B. Equally
      C. In the same proportions to current accounts
      D. Equally after adjusting capital balances
      E. As required by law of the country.

   iv. A provision for bad debts is created when.
      A. Debtors becomes bankrupt.
      B. Debtors cease to be in business.
      C. In need of creating bad debts.
      D. In need of writing off bad debts.
      E. Debtors becomes absolute.

   v. Which of the following is nominal account?
      A. Bank A/C B. Furniture and fittings.
      C. Motor vehicles A/C. D. Motor vehicles running expenses.
      E. Bank overdraft.

   vi. Which document is raised when goods are received from debtor?
      A. Debtor note B. Credit note
      C. Goods received note D. Sales invoice.
      E. Purchases invoice.

   vii. Which of the following should be charged to profits and losses A/C?
      A. Work in progress
      B. Direct materials costs
      C. Carriage on raw materials

D. Office rent
E. Indirect labour.

viii. Receipts and payments account does not show.
A. Cheque paid out during the year
B. Accumulated fund
C. Receipts from sales of fixed assets
D. Bank balance
E. Cash balance.

ix. Which of the following is correct?
A. Drawing increases capital.
B. Profits decreases capital.
C. Loss increases capital.
D. Drawings are added in capital
E. Profits increases capital.

x. Which of the following is not an asset?
A. Loan to somebody.
B. Bank balances
C. Bank over draft.
D. Cash borrowed from somebody.
E. Furniture bought on credit.

2. Match the responses in **list A** with the statements in **list B** by writing the letter of the correct response beside the item number.

LIST A	**LIST B**
(i) Is the revenue received in advance.	A. Error of omission.
(ii) Is when a transaction is completely forgotten form the books of accounts.	B. Error of commission.
(iii) Is when the double entry observe the incorrect figures.	C. Error of principle.
(iv) Is a temporary account used to make the trial balance totals equal.	D. Compensating error
(v) Is when the correct amount is entered in the wrong class of account.	E. Error of original error
(vi) Is the revenue which is not yet received.	F. Error of completely reversal of entries.
(vii) Is when the correct amount is entered in the wrong personal accounts	G. Correction of errors.
(viii) Is the process by which mistakes committed	H. Suspense account.
	I. Accrued revenue.
	J. Unearned revenue.

	in the books of accounts are corrected.	
(ix)	Is when the error cancel out each other.	
(x)	Is when the transactions are shown in wrong side of the account.	

SECTION B (20 Marks)

Answer **all** questions in this section.

3. From the following you are to draw up the trading account for Fine's Department Store for the year ended 30 June 2016.

Stocks: 1.7.2015 30.6.2016

	Tshs.	Tshs.
Carpet Department	16,100	18,410
White Goods Department	37,916	35,119
Music Department	31,222	40,216

Sales for the year:	**Tshs.**
Carpet Department	62,400
White Goods Department	151,300
Music Department	94,820

Purchases for the year:	
Carpet Department	43,600
White Goods Department	118,260
Music Department	55,924

4. (a) Give out five (5) factors that can cause a difference in balance between cash book and bank statement.

(b) List three purposes of a cash book.

SECTION C (60 Marks)

Answer **all** questions in this section.

5. (a) As a business student with basic knowledge of Accounting identify possible mistakes from the following statements and correct them via Journal Entries.

   i. Cash sales of Tshs. 30,000 were debited Bank.
   ii. Cash paid for expenses were debited to Expenses.
   iii. An amount of debtors balance Tshs. 10,000 was not included in Debtors Account.
   iv. Purchases of Building Tshs. 40,000 were recorded to Buildings Account.
   v. Discount allowed was undercasted by Tshs. 200.

   (b) Prepare Partners Appropriation Account from the following information extracted from Abdul-aziz & Alex Partnership who shares Profits and Losses equally:

Gross income less expenses of the business			13000
Salaries of the partners:	Abdul-aziz	1000	
	Alex	1000	
Commissions:	Abdul-aziz		1500
	Alex	Nil	
Bonuses to partners:	Abdul-aziz		Nil
	Alex	1500	
Interest on Capital:	Abdul-aziz	30000	
	Alex	29000	
Interest on Drawings:	Abdul-aziz		1000
	Alex	600	

6. At the beginning of the financial year on 1 April 2015, a company had a balance on plant account of Tshs.372, 000 and on provision for depreciation of plant account of Tshs.205, 400.
The company's policy is to provide depreciation using the reducing balance method applied to the fixed assets held at the end of the financial year at the rate of 20% per annum.
On 1 September 2005 the company sold for Tshs.13, 700 some plant which it had acquired on31 October 2001 at a cost of Tshs.36, 000?

The company acquired new replacement plant on 30 November 2005 at a cost of

Tshs.96, 000, inclusive of installation charges of Tshs.7, 000.

Required:

Calculate:

(a) The balance of plant at cost at 31 March 2016

(b) The provision for depreciation of plant at 31 March 2016

(c) The profit or loss on disposal of the plant.

7. (a) Use you comprehensiveness ability from single entry to calculate a Gross Profit for the year ending 2015, data bellow were made available:

Account Name	**Balance (Shs.)**	
	2017	**2018**
Stock	35000	28000
Debtors	400000	650000
Creditors	350000	470000
Cash Transactions		**Shs.**
Sales		300000
Purchases		195000

(b) Stanley and Barclay enter a joint venture to share profits or losses equally resulting from dealings in second hand digital TVs. Both parties take an active part in the business, each recording his own transactions. They have no joint banking account or separate set of books.

2018

July 1 Stanley purchased four TVs for a total of Shs.1, 100.

3 Stanley pays for repairs Shs.840.

4 Barclay pays office rent Shs.300 and advertising expenses Shs.90.

6 Barclay pays for packaging materials Shs.34.

7 Barclay purchaseds a TV in excellent condition for Shs.600.

31 Stanley sells the five TVs to various customers, the sales being completed on this date, and totaling Shs.3, 100.

Show the relevant accounts in the books of both joint ventures and memorandum joint venture account.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

032/1 **CHEMISTRY 1**
(For Both School and Private Candidates)

Time: 3Hours **Thursday, 27th June 2019 a.m.**

Instructions

1. This paper consist of sections A, B and C with total of **thirteen(13)** questions
2. Answer **all** questions in this paper.
3. Calculator, cellular phones and any unauthorized material are **not** allowed in the examination room.
4. Write your **Examination Number** on every page of your answer booklet(s).
5. The following constant may be used.

   Atomic masses:
   H = 1, Li = 7, C = 12, O = 16, Na = 23, Al = 26, Cl = 35.5, K = 39, Ag = 108
   Avogadro's number = 6.02×10^{23}
   GMV at s.t.p. = 22.4 dm^3 .
   1 faraday = 96,500 coulombs.

Standard pressure = 760 mm Hg.

Standard temperature = 273 K.

1 litre = 1 dm^3 = 1000 cm^3 .

SECTION A (20 marks)

Answer **all** questions in this section.

1. For each of the items (i) (x), choose the correct answer among the given alternatives and write its letter beside the item number.

(i) The mass number of a carbon atom that contains six protons, eight neutrons, and six electrons is

A 6 B 14 C 8
D 12 E 20

(ii) How many moles of oxygen are required for the complete combustion of 2.2 g of C_3H_8 to form carbon dioxide and water?

A 0.050 moles B 0.15 moles C 0.25 moles
D 0.50 moles E 0.025 moles.

(iii) Which of the following hydrocarbons does NOT belong to the same homologous series as the others?

A CH_4 B C_3H_8 C C_4H_{10}
D C_6H_{12} E C_2H_{12} .

(iv) Chemical change means;

A The change is reversible. B Can easily be separated.
C The change is complete. D New substance is produced.
E Produces no change of mass.

(v) 10 cm3 of 0.4 M sodium hydroxide are added to 40 cm3 of 0.2 M hydrochloric acid. The resulting mixture will be

A Neutral B Alkaline C Dilute
D Acidic E Amphoteric

(vi) Consider the system at equilibrium: $H_2O(l) \rightleftharpoons H_2O(g)$ for which $\Delta H > 0$. Which change(s) will increase the yield of $H_2O(g)$.

A Increase in temperature
B Increase in the volume of the container
C Increase in temperature and volume of the container
D Increasing surface area of oxygen
E Increasing surface area of reactants.

(vii) Which of the following compounds contains only two elements?

A Magnesium hydroxide B Magnesium nitride
C Magnesium phosphate D Magnesium sulphite
E Magnesium sulphate.

(viii) Which one of the following sets of laboratory apparatus are used for measure volume?

A Crucible, U-tube and volumetric flask
B Test tubes, beakers and glass jar
C Thistle funnel, separating funnel and beaker
D Burette, pipette and measuring cylinder
E Conical flask, test tube and measuring cylinder.

(ix) _____ is the general term used to explain a mixture of different metals

A Alloy B Allotrope C Amphoteric
D Amorphous E Isotope.

(x) A solvent can be obtained from a solution by

A evaporation followed by decantation B filtration and condensation
C evaporation and filtration D evaporation and condensation
E crystallization followed by sublimation

2. Match the items in **List A** which the responses in **List B** by writing the letter of the correct response beside the item number.

LIST A	**LIST B**

i.	Its carbonate is used to remove hardness of water.	A. Oxygen B. Carbon dioxide
ii.	Used in the manufacture of ammonia.	C. Chlorine
iii.	It is the strongest oxidizing agent among the	D. Carbon
iv.	halogens.	E. Fluorine
v.	Its chloride is added to food in order to give taste.	F. Hydrogen G. Nitrogen
vi.	It exists in three main forms.	H. Calcium
vii.	A group of atoms with unpaired electrons	I. Potassium
viii.	Vital for all living things	J. Radical
ix.	Acid rain	K. Electron
x.	Inorganic chemical compounds which are added to the soil to supply more nutrient.	L. Sulphur Dioxide M. Fertilizer
xi.	Incredibly stable and rarely reacts.	N. Noble Gases O. Sodium

SECTION B (54 Marks)

Answer **all** questions in section

3. (a) Study the following portion of the periodic table with some elements represented by letters and answer the questions that follow

I							VIII
	II	III	IV	V	VI	VII	
						B	
	A				C	D	

(i) State how electro negativity varies from A to C and from B to D

(ii) Write the electronic configurations of A, C^{2-}, D and B.

(b) Consider elements with atomic number 1, 11, 12 and 17

i. What are the types of oxides formed by elements with atomic number 11 and 12?

ii. Write an equation which represents a reaction between the element with atomic number 1and 17.

iii. Write a balanced chemical equation between the oxide of the

element with atomic number 11 and aqueous solution of the compound formed in 3 (a) (ii).

4. (a) Giving four reasons, explain why people who use hard water can expect higher costs than people who use soft water

(b) Suggest one method for the separation of each of the following

i. Iodine and sand.
ii. Green solution from leaves.
iii. Alcohol and water.
iv. Iron fillings and powdered calcium carbonate

5. The following are the general structural formulae of certain organic compounds:
R-OH, R-COOH, and RCOOR'.

(a) Name the:

i. Homologous series represented by R-OH, R-COOH and RCOOR'.
ii. Functional groups represented by R-OH and R-COOH.

(b) When a burning splint is introduced into a gas jar containing carbon dioxide, the flame goes out.

i. What two properties of carbon dioxide does this experiment illustrate?
ii. What type of equipment used widely in everyday life makes use of these two properties?

6. (a) Give three examples in each of the following:

i. Solid fuel
ii. Gaseous fuel

(b) The reaction which produces methanol from carbon monoxide and hydrogen is represented by the equation $CO(g) + 2H_2(g) \quad CH_3OH(g)$ $\Delta H = -94kJ\ mol$ The reaction is carried out at high pressure to give a good yield of methanol.

i. Explain why increase in pressure gives a better yield of methanol.
ii. The value of ΔH is negative. What does this tell about the reaction?
iii. With a reason, state whether a high temperature or low temperature will give better yield of methanol.

7. (a) Give the names or formula of the two chemicals that would be used in the laboratory to make each of the following gases. State a simple test that could be used to identify each gas.

i. Oxygen.
ii. Hydrogen.
iii. Carbon dioxide

(b) Suggest a suitable indicator for the following titrations:
   i. Hydrochloric acid against ammonia solution.
   ii. Sulphuric acid against sodium hydroxide solution.
   iii. Ethanoic acid against potassium hydroxide solution

8. (a) 25 cm^3 of 0.1 M HCl were neutralized by 23 cm^3 of sodium hydroxide solution.
   Calculate the concentration of the alkali in grams per litre.

   (b) Give the meaning of the following terms:
   i. Soil structure.
   ii. Acidic soil.
   iii. Liming.

9. (a) Current of 0.5 A were made to flow through silver voltammeter for 30 minutes.
   Calculate the mass of silver deposited and the equivalent weight of silver.
   (b) Explain the following reactions giving one example in each:
   (i) Addition reaction.
   (ii) Elimination reaction

10. (a) (i) Name three gases which should not be produced in order to prevent the
   destruction of ozone layer
   (ii) List and explain three effects of ozone layer depletion.

(b) Lack of safe water for domestic and industrial uses is a serious problem in most of Tanzanian towns. The major cause of this problem is pollution in the water sources. Slate three methods that could make water from a pond or a well be safe for drinking.

11. In the blast furnace, iron ore can be reduced using coke at a temperature of about 1300°C.
   (i) Write an equation for the exothermic reaction that causes this high temperature.
   (ii) State how carbon monoxide is formed.
   (iii) Write a word equation for the formation of slag.

SECTION C (26 Marks)

Answer **all** questions in this section.

12. Consider a four carbon hydrocarbon (C4Hn), where n is an integer. Give the

name of homologous series, molecular formula and structural formula for different isomers of the compound formed by each homologous. In each case indicate the causes of isomerism.

13. Addition of inorganic fertilizers in the farm is not as important as addition of organic manure. Discuss the correctness of this statement in four points.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

011 **CIVICS**

(For Both School and Private Candidates)

Time: 2:30Hours **Monday, 24th June 2019 p.m.**

Instructions

1. This paper consists of sections A, B and C with a total of **ten (10)** questions.
2. Answer **all** questions in section A and B and **three** questions from section C.
3. Section A and B carry 20 marks each and section D carries 60 marks.
4. Cellular phones and any unauthorized material are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s).

SECTION A (20 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter beside the item number in the answer booklet(s) provided.

(i) Tanganyika became a republic on

A 9th December 1961 B 9th December 1962 C 26th April 1964

D 12th January 1964 E 14th October 1963

(ii) Which one of the following is a characteristics of formal sector in Tanzania

A Employees wages are often low
B Lack of permanent business places
C Employees lack social security protection
D Low capital investment
E Business activities are regulated by taxation law

(iii) The current constitution of the United Republic of Tanzania was enacted in

A 1977 B 1961 C 1964 D 1964 E 1965

(iv) Which of the following is the second step in problem solving process?

A Initiating mediation
B Agree in solution
C Gathering information
D Defining the problem
E Generating alternatives

(v) The ability of a person to think and come up with new idea is known as

A Creative thinking
B Critical thinking
C Decision making
D Problem solving
E Deep thinking

(vi) The major driving force of globalization is

A Rapid population growth
B Revolution in communication and information technology
C Improvement of transport and communication
D Development of multi-lateral co-operation
E Colonialism

(vii) The way male and female are portrayed in the society is referred to as
A Gender Blind B Gender discrimination C Gender role
D Gender stereotyping E Gender balance

(viii) The parliament of the United Republic of Tanzania is divided into ;
A The prime minister and the National Assembly
B The president and the National Assembly
C The speaker and the National Assembly
D The National Assembly and the Parliament
E The Clerk and the National Assembly

(ix) The following are indicators of poverty except
A low per capita income
B Low level of science and technology
C Low rate of infant mortality rate
D Low life expectance
E High illiteracy rate

(x) Aspect of culture which involves the tangible things
A Non-material culture B Traditions C Customs
D Language E Material culture

2. Match the items in List A with the correct response in List B by wring the letter of the corresponding response beside the item number in the answer booklet(s) provided.

List A	List B
i. The exercise of fairness and justice to all sexes without discrimination. ii. The situation where one sex is treated either better or worse than the other. iii. The way men and women are shown in the society. iv. Inability to know that there are different gender roles v. The systematic effort to identify and understand the roles of women and men within a given society.	A. Gender analysis B. Gender blindness C. Sexuality D. Female Genital Mutilation E. Gender equity F. Gender discrimination G. Gender balance H. Gender equality I. Widow inheritance J. Gender stereotyping

vi. A globally accepted strategy for promoting gender equality. vii. Social relationship between male and female. viii. Biological different between male and female. ix. Situation where by women are treated badly compared to men. x. The removal of the outer part of the sexual organ of a women	K. Bride price L. Sexual intercourse M. Women discrimination N. Gender oppression O. Women empowerment P. Sex Q. Gender R. Gender mainstreaming S. Polygamism

SECTION B (20 Marks)

Answer **all** questions in this section

3. Read the following passage and answer the questions that follow.

Many people participate in wealth production in our country. Tanzania is endowed with many resources. It has fertile land and favorable climate. However, labour is needed in order to benefit from these favorable factors. In our country labour sources include; peasants, small scale farmers, livestock keepers, traders, workers in crafts, and industrial workers in general. Most of working force is employed in agriculture about 75 -80 percent of Tanzania working force is employed in agriculture. This is because our agriculture is not mechanized. The majority of agricultural producers in our country are peasants. They still use simple tools such as hoes.

In addition to farming, our country is rich in livestock consist of cattle, goats, sheep, poultry and pigs. Livestock keepers provide Tanzania with meat, milk and eggs which are very important a source of protein. They are also source of government revenue, for example hides and skins are exported to earn foreign currency. They are also used locally in making shoes for people.

Another important activity and source of employment and income is fishing. Much labour is used in fishing especially for those living near lakes and oceans. Unfortunately, the fish industry is seriously affected by lack of experts and efficient modern fishing technology. Some fishermen use dangerous and destructive methods such as explosive in fishing.

Questions

(a) Identify the suitable title of this passage.

(b) According to the passage, what are four main sources of labour in Tanzania?

(c) With reference to the passage, how can Tanzania benefit from livestock keeping sector?

(d) List down two problems that hinder fishing industry in Tanzania.

(e) In your own words, what can be done in order to improve the agriculture sector in Tanzania? (Give four points).

4. (a) (i) What is local government?

(ii) List down four sources of local government revenue.

(b) Outline five importance of national constitution.

SECTION C (60 Marks)

Answer **three (3)** questions in this section. Each question carries **20** marks.

5. Examine six cause root of improper behavior in a society.
6. "Globalization is blessing to Tanzania." Defend this statement by using six points.
7. Examine six advantages of the private sector to the economy of Tanzania.
8. "Failure to apply life skills can lead to problems." Identify six consequences of not applying life skills.
9. Tanzania is credited for promoting and broadening the scope of democracy. Give six points to show the validity of this statement.
10. "To a great extent, the road accidents in Tanzania are manmade." Justify this statement by giving six points.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

061 **COMMERCE**
(For Both School and Private Candidates)

Time: 2:30 Hours **Tuesday, 2nd July 2019 a.m**

Instructions.

1. This paper consists of section A, B and C with a total of **ten (10)** questions.
2. Answer **all** questions in section A and B and **two (2)** questions from section C.
3. Calculators and cellular phones are **not** allowed in the examination room.
4. Write your **Examination Number** on every page of your answer booklet(s).

SECTION A: (20 Marks)

Answer **all** questions.

1. For each of the items (i) – (ii), choose the correct answer from among the given altenatives and write its later beside the item number in your answer sheet.

   (i) Which of the following document must be complited by an importer when clearing goods before custom officer?
   - A. Bill of sale
   - B. Bill of lading
   - C. Bill of entry
   - D. Bill of sight
   - E. Warehouse warrant

   (ii) In hire purchase buying:
   - A. Hired cars are bought
   - B. Sellers remains the owner of the goods until payments of the last installment.
   - C. Buyers own the goods on paying the first installment
   - D. Purchases are made from hire premises
   - E. Seller remains the owner of the goods regadless the payment of all installment

   (iii) The basic commercial activities are:
   - A. Advertising, banking and transport
   - B. Export trade, import trade and retail trade
   - C. Warehousing, insurance and transport

D. Retail trade and wholesale trade
E. Home trade and foreign trade

(iv) PAYE is a tax:
A. Imposed on incomes of individuals
B. Imposed on income of employees
C. Levied on profits of limited companies
D. Imposed on expensive imported goods
E. Imposed on the sale of stocks

(v) If goods were listed at 20,000/= how much would be paid by customer if he was allowed 10% trade discount and 2% cash discount?
A. 17,650/=
B. 17,460/=
C. 17,640/=
D. 17,560/=
E. 17,820/=

(vi) Marketing of imported and locally manufactured goods in Tanzania is largely under:
A. National Distribution Limited (NDL)
B. Board of External Trade (BET)
C. Board of Internal Trade (BIT)
D. Cooperative societies
E. Board of Directors

(vii) The main objective of commerce is to:
A. Enable consumers to buy goods
B. Enable traders to earn profit
C. Bridge the gap between producers and consumers
D. Help manufacturers to produce goods for use
E. Satisfy human needs

(viii) A favourable balance of trade means:
A. A country's value of exports exceed its value of imports within a specific year.
B. A country's value of imports exceed the value of exports within a specific year.
C. Excess of the growth of the value of goods produced in a specific year compared with the previous year
D. More traders have moved in the country in the specific year compared with the previous year.
E. More goods were registered to be exported in the specific year than in the previous year.

(ix) Profoma invoice is:
A. An inquiry
B. An order

C. A quotation
D. A receipt
E. A cheque

(x) Which of the following items does not appear in the memorandum of an association
A. Power of directors
B. Type and amount of capital
C. Objects of a company
D. Name of company
E. Liability clouse

2. Match the responses in LIST B with statements in LIST A by writing the letters of the correct response beside the item number.

LIST A	**LIST B**
(i) Is the whole process of putting or arranging goods manufactured by a firm	A. Advertising variables B. Markerting mix
(ii) The exclusive legal rights to reproduce, publish and sell the product in the form of literacy, musical or artistical work	C. Variable grading D. Fixed grading E. Marketing research
(iii) Market structure with few suppliers and many buyers	F. Product investigation G. Export merchants
(iv) This is the part of brand which can be pronounced when a buyer order for the product	H. Board of External Trade I. Monopolistic market
(v) Set of controllable variables that the firm can use to influence the buyers response.	J. Oligopoly K. Perfect compitition market
(vi) A situation under which rival firms undertake a series of price reduction with the aim of cupturing a greater market.	L. Brand mark M. Brand name
(vii) Refers to the process whereby the same standards are used year after year.	N. Price freedom O. Price war
(viii) An investigative activity which is carried out to establish the consumers demand for a product	P. Packaging Q. packing
(ix) Undertaking marketing goods for exports	R. Producer's right
A market structure where there is many buyers and sellers with homegeneous product sold in the market	S. Copy rights

SECTION B (40 Marks)

Answer **all** questions in this section

3. Write is meant by the following insurance terms
   a) Double insurance
   b) Re-insurance
   c) Cover note
   d) Assessors
   e) Premium

4. (a)Outline five features of a good tax system
   (b) In briefly analyze five merits of direct taxes

5. Differentiate the following terms
   a) Direct production and indirect production
   b) Chain store and departmental stores
   c) Bill of lading and charter party
   d) Share and debenture

6. Study the schedule below:

Demand in kg	200	400	600	800	1000
Price in shs per kg	100	80	60	40	20
Supply in kg	1000	800	600	200	100

   a) From the above information calculate the price elasticity of demand when price changes from shs. 40 per kg to shs. 20 per kg.
   b) State whether the price elasticity of demand is elastic or inelastic or unitary. Give reasons.

SECTION C (40 Marks)

Answer **two (2)** questions from this section

7. (a) Explain five features of a private limited company.

   (b) What is a partnership deed? Outline five contents of a partnership deed.

8. Examine six services offered by Tanzania Postal Office.

9. Discuss current trends in Tanzania banking and financial services.

10. (a) What is containerization?
    (b) Explain five advantages and three disadvantage of containerization.

015/1 **ELIMU YA DINI YA KIISLAMU 1**

(Kwa watahiniwa Walioko na Wasiokuwa Shuleni)

Muda: Saa 2 **Jumamosi, 29 Juni 2019**

mchana

Maelekezo

1. Karatasi hii ina sehemu A, B, C , D na E zenye jumla ya maswali matano (5)..
2. Jibu maswali **yote..**
3. Simu za mkononi na vitu vyote visivyoruhusiwa **havitakiwi** katika chumba cha mitihani..
4. Andika **Namba yako ya Mtihani** katika kila ukurasa wa kijitabu chako cha kujibia.

SEHEMU A: (Alama 10)

SUNNA NA HADITHI

1. Chagua jibu sahihi zaidi kwenye kipengele (i) – (x) uandike herufi yake kwenye kitabu cha majibu.

   i. Imam Li Allah amridhiye aliandika kitabu mashuhuri kiitwacho
      A. Al-muwatta
      B. Ar.risaala
      C. Al-wajid
      D. Wasilatu shafii
      E. Al;Qatadaya

   ii. Hadithi inayokubalika kuwa iko kwenye daraja la kwanza kwa usahihi inapatikana kwenye vitabu vya
      A. Al-muwatta
      B. Al-qatadaya
      C. Bukhar na Muslim
      D. Bulughul-marami
      E. Sanan Abi Daud na Tirmidh

   iii. Fani inayomjulisha msimulizi wa hadithi kuwa mkweli au simkweli huitwa
      A. Nas-khuna Isnaad
      B. Jar-huwataadil
      C. Taadil na Hasan
      D. Isnaad na Rawi
      E. Bukhar na Muslim.

   iv. Aina kuu za Isnaad kwa mujibu wa elimu ya Hadithi ni
      A. Ash-hur na Azizi
      B. Mualaqa na Gharib
      C. Muttasil na Muarsal
      D. Muttawattir na Ahad
      E. Muudal na Ahad.

   v. Amesimulia Abdallah bin Yusufu toka kwa Malik Toka kwa Abi zaidi toka kwa A'raji toka kwa Abu Hurairat ziwe juu yake …(Bukhar). Huu ni mfano mzuri wa
      A. Hadithi
      B. Isnaad
      C. Isnaad na Mtini

D. Matini na Rawi
E. Riwaya

vi. Kipindi cha karne ya pili AH 101 – 200 AH kulijitoeza mwanachuoni mwashuhuri akaandikakitabu na kikawa ni kimbilio la wanachuoni wakati huo na chenye msaada mpaka leo; huyo ni Imam

A. Muhammad bin Idrisa-As-Shaafi Allah amrehemu.
B. Abu Hanifa Allah amrehemu
C. Bukhar Allah amrehemu
D. Muslim bin Hajjaaj Allah amrehemu
E. Malik bin Anas Allah amrehemu

vii. Katika fani ya hadithi neno "Mauquf" lina maana ya
A. Isnaad imekatikia kwa maswahaba
B. Simulizi na taarifa za maswahaba
C. Taari kutoka kwa Taabiina
D. Isnaad ya hadithi haifiki kwa Mtume sala na salamu ziwe juu yake.
E. Hadithi ambayo imesimuliwa na watu wawili

viii. Hadithi ahad ni hadithi ya aina gani?
A. Ni hadithi ambayo haina Isnad sahili
B. Hadithi ambayo inamatatizo ndani yake
C. Hadithi ambayo Mtume sala na salamu amesisitiza jambo fulani
D. Hadithi ambayo haipingani na Qur'ani
E. Hadithi amabyo imepokelewa na mpokezi mmoja

ix. Ili hadithi ikubalike kwa wanachuoni wa hadithi, ni lazima yapatikane mambo yafuatayo:-
A. Naswu na Riwaya
B. Sahihi na Hasan
C. Isnaad na matini
D. Jar-hu na Muslimu
E. Imam Suyuti na Bukhar

J. Hadithi inayosimuliwa na wasimuliaji wengi kupitia riwaya mbalimbali inajulika kwa jina la :-
A. Ahad
B. Sahihi
C. Mutawattir
D. Hasan
E. Maudhui au dhaifu

SEHEMU YA QUR'AN (Alama 15)

2. Oanisha neno kutoka **ORODHA "B"** linalokubaliana na neno kutoka **ORODHA "A"** Herufi ya jibu liandike kwenye kitabu chako cha majibu.

ORODHA "A"	ORODHA "B"
i. Na wasiohukumu kwa yale aliyoteremsha Allah mtukufu basi ni makafiri (5:44)	A. Njia anazozitumia Allah kuwasiliana na waja wake
ii. Nawanapoambiwa msifanye uharibifu ulimwenguni husema "sisi ni watengenezaji" (3:11 - 13)	B. Madai ya Makafiri
iii. Soma kwa jina la Mola wako aliyeumba . (96 :1 - 5)	C. Suratul Takaathur
iv. Na walikaa pangoni miaka miatatu na kenda (18:25).	D. Suratul Tiin
v. Na ikiwa mnashaka na kile alicho teremshiwa mja wetu basi leteni mfano wake ….."	E. As-haabul ukhudud
vi. J e ? Allah si Hakimu Mwadilifu? Kwa njia ya Il-ham, Wahyi ……" (42:51).	F. As-haabul Kahfsuratu – Maaidah
vii. Kasha kwa hakika mtaulizwa siku hiyo juu ya neema (102 : 07)	G. Sababu za kuingia motoni
viii. Atiyaye wasiwasi nyoyoni mwa watu (114:5)	H. Wanafiki
ix. Ninyi mnadini yenu na mimi nina dini yangu (109 : 6)	I. Amri ya kwnza ni kusoma
x. Na kama wakikutaka amani; wape amani (8 : 61)	J . Iblis Alaih laana
xi. Mtume sala na salamu ndiye kigezo cha uislamu	K. Siku ya Qiyama
xii. Sal;man Rushd	L. Maisha ya Kaburini
	M. Ukafiri nao ni dini
	N. Qur'an ni njozi za Mtume sala na sakamu juu yake zilizovurugika
	O. Uislamu unahimiza amani
	P. Suratul Baqarah aya ya 23
	Q. Qur'an ilishuka kwenye pangola Hira
	R. Mkataba wa Hudaibyah
	S. Vita vya Uhud
	T. Bilashaka mnamfano mwema kwa aaminiye Allah na siku ya mwisho
	U. Msafara wa Tabuk

xiii.	Anderson na watoto	
xiv.	Bila shaka tumekupa kushinda kuliko dhahiri.	

SEHEMU C (Alama 15)

TAWHID

3. Panga sentensi katika vipengele vya (a), (b) na (c) katika mtiririko wenye kuleta mantinki. **Andika herufi na namba tu** zinazohusika katika kijitabu chako cha kujibia.

(a) Baadhi ya Tafsiri ya Suratul Humazah

i. Ambao unapanda nyoyoni
ii. Kwa magogo marefu marefu
iii. Na nijambo gani litakujulisha nini hiyo Hutamali?
iv. Hakika huo moto watafungiwa
v. Ni moto wa Allah mtukufu uliowashwa barabara.

(b) Allah azza wajalla anawataka waja wakumbuke mambo mbalimbali kwenye sura ya (78:6-10)

i. Na tukafanya usiku kama nyuo
ii. Na milima kuwa kama vigingi
iii. Na usingizi wenu kama kifo
iv. Na tukakuumbieni wanaume na wanawake
v. J e hatukuifanya ardhi kuwa tandiko?

(c) Kwa mujibu wa hadithi mambo tutakayoulizwa siku ya kiyama nikama ifuatavyo: -

i. Umeitumiaje elimu yako?
ii. Umeutumiaje umri wako?
iii. Mali yakoumeichumaje na umeitumiaje?
iv. Ujana wako umeutumiaje?
v. Mwili na vipaji vyako umevitumiaje?

SEHEMU D: FIQHI (Alama 30)

3. J ibu maswali yafuatayo kwa kifupi kwa kuandika majibu yako katika kijitabu chako cha kujibia.

a) Ainisha mambo manne (04) ambayo yanataka mja katikaswala.

b) Kwa mujibu wa (2;184); taja makundi ya watu wawili yasiyowajibika kufunga Ramadhani
c) Katika kufunga ndoa neno :Nimekuoza: linaitwa vipi na neon "Nimekubali"linaitwa vipi?
d) Taja misingi mitatu ya sheria za kiislamu
e) Maji masafi yatafaa kwa twahara yakiingia najisi isipokuwa yakigeuka moja ya sifa tatu zitaje
f) Miongoni mwa majosho ya Sunna ni kuoga kwa ajili ya swala za jamaa. Taja mengine matatu
g) Taja nguzo nne (04) za kutamka ndani ya swala
h) Eleza juu ya talaka khulii inavyokuwa
i) Mwanamke aliyefiwa na mumewe anakaa siku ngapi eda?
j) Eleza kwa ufupi ukamilifu wa sanda ya mwanamke
k) Ili mwizi akatwe mkono ni lazima sharti zikamilike zzitaje tatu
l) Ni ipi hukumu ya mwislamu aliyeritadi (aliye uacha Uisalmu)?
m) Taja viwango vya zaka ya Ngamia na mbuzi
n) Bainisha matendo manne (04) yanayofanyika mwenzi 10 dhul-Hija
o) Eleza zakat ya Mali ya kuokota nay a kuchimbuliwa ardhini

SEHEMU E: (Alama 30)
TAREKH

4. Chagua **neno** au **vifungu vya maneno** (i) – (xxx) ili ukamilishe sentensi (a) – (o) **kwa kuandika majibu tu** sahihi katika kitabu cako cha kujibia. Maneno au vifungu vya maneno vinaweza kutumika zaidi ya mara moja.

(i) Abu J ahal
(ii) Qur' an
(iii) Othman bin Affan Allah amridhiye
(iv) Yathrib
(v) Kaab bin Malik
(vi) Wajemi na warumi.
(vii) Lulu Al- majusi (Fizoe)
(viii) Daud Alaih salamu
(ix) Abu lahab
(x) Uhud
(xi) Haruna Alaih salamu
(xvi) Alama za kuwepo Allah
(xvii) Abdul-rahman bin Muljam
(xviii) Peponi
(xix) Ufasaha
(xx) Baraka bin Abdallah
(xxi) 309
(xxii) 132 miaka
(xxiii) Yusufu Alaih salam
(xxiv) Abu Ayoub Al-answer
(xxv) Hudaibiya
(xxvi) Muhammad sala na

salamu

(xii) Asma Allah amridhiye
(xiii) Jabal safaa
(xiv) Ghaari Thauri
(xv) Adam na Hawa Alaihima salam
(xxvii) Yarmuk
(xxviii) Nuhu Alaih salam
(xxix) Mtume
(xxx) Mtu
(xxxi) Umar bin khatab Allah amridhiye
(xxxii) Abu sufyan
(xxxiii) Joseph

MASWALI

a) Bibi _____ ni mtu ambaye alikuwa anawahudumia Mtume sala na saamu na Abu Bakar Allah amridhiye kwenye _____
b) Kiongozi wa vita vya ___alikuwa _____
c) Swahaba _____alipopigwa jamabia ikiwa ndio sababu ya kifo chake; alipigwa na _____
d) Nabii Adam Alaih salam ni _____na _____ wa kwanza hapa duniani.
e) Allah Azza wjallah alimkubalia Nabii Mussa alaih salaam kwa kumfanya ndugu yake ___ awe ni msaidizi wake wa da'awa kwa _____ wa kuongea
f) Kwenye mkataba wa _____Mtume sala na salamu alimtuma _____ kwa makuraishi kuwajulisha dhamira yao ya kwenda makka
g) Mtume sala na salamu alipopewa amri ya kulingania watu kwa dhahiri alipanda jabali _____ na miongoni mwa jamaa zake waliokuwepo hapo ni _____
h) Upinzani kati ya ___ulirudi tena katika kipindi cha _____
i) Khalifa Ali bin Talib alipangiwa auliwe na _____ na Mwawiya bin Abi sufyan auliwe na _____
j) Nabii _____ pia aitwa _____
k) Watu wa pangoni walikaa humo miaka _____ na aliteuliwa kuwa waziri katika serikali ya Farao.
l) Nabii _____ ni mtoto wa Nabii _____ ambaye alihukumu kesi iliyomshinda baba yake.
m) Mtume sala na Salamu alipofika ____alifikia kwa _____
n) Sababu kubwa ya mtume Muhamammad sala na salam ziwe juu yake kwenda Isra na Miraji imebainishwa katika _____ ambayo ilikuwa na dhumuni la kuoneshwa_____
o) _____ walianza maisha yao _____.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

022 ENGLISH LANGUAGE
(For Both School and Private Candidates)

Time: 3 Hours Thursday, 27th June 2019 p.m.

Instructions

1. This paper consists of sections A, B, C and D.
2. Answer the questions as instructed under each section.
3. Cellular phones are **not** allowed in the examination room.
4. Electronic calculators are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s).

SECTION A (10 Marks)

COMPREHENSION AND SUMMARY

Answer **all** questions in this section.

1. Read the passage below carefully and answer the questions that follow

There are three kinds of education in Africa. There is the old traditional education; there are the remains of the colonial schooling, which varied according to how the European power saw African requirements; and there is the post-independence attempt to find an education suitable for the needs of modern Africa.

The old education grew naturally out of the village and tribal ways of life, and there was much to commend it. A child had to learn how to deal with the dangers of his surroundings and how to treat his fellows. He knew about weather, love and the skills of a spear, axe and hoe from old men. His mother taught him correct speech, behaviour and respect for his elders. Throughout his childhood, it was impressed on him that he could not live alone - he must 'conform', and accept the ways of the tribe. He lived in a world of kinship: his kin gave him security in sickness and old age. Law and ownership of land, too, were based on kinship. So the child learned the rites which kinship would demand throughout his life from birth to maturity, marriage and death.

Much of this education was informal. The home was the child's school, where he learned traditional legends and proverbs. (For example, the reader might like to work out the meaning of these two Baluyia proverbs from Kenya: 'A person running alone thinks he is the fastest runner' and 'a small bird cannot advise a bigger one). This social education had a great emphasis on correct conduct and confidence. A traveller in Africa in 1930s wrote: 'I have seen three children between four years and six quite competently preparing a meal with no supervision'. But severe tests of endurance were a more formal part of this education. Chagga boys in the old days had to sleep in holes in the ground at night for nine months, often in the cold mountain air. A boy was required to go on a lonely expedition into the forest to kill a leopard with a bow and arrow. Bena girls aged nine to thirteen, were ducked repeated in streams, or terrified by women pretending to fall dead at their feet, and by the appearance of monsters.

This traditional form of education had the advantage of preparing a child for life in the community; it did not in general encourage him to be ambitious or independent, or teach him to meet the needs of the modern world. So in the colonial years

missions and 'European' schools taught the kinds of things children in Europe were taught. This produced small westernised elite in some colonies, but it was severely criticised for having little to do with African needs. Reading books had European birds and snow scenes in them; arithmetic problems dealt with taps, and wall-paper rolls. Few of these things had anything to do with many African children's environment.

Colonial education too has been criticised because it was, to quote President Nyerere of Tanzania, 'motivated by a desire to inculcate the values of the colonial society andto train individual for the service of the colonial states'. The state interest in education was based on the need for local clerks and junior officials, and there was thus a heavy emphasis on subservient attitudes and white-collar skills.

In the post-independence era, African needs are being rethought. Questions are being asked such as, how many universities should a country have, whilst half of an age group gets no schooling at all? Some highly trained electrical engineers are needed for the power stations; but Africa also needs men skilled in the relatively simple skills of wooden bridge construction, laying late rite roads and building single-story houses. Civil servants especially need a good secondary education if they are to deal with matters ranging from money for a new agricultural scheme to collecting information for government approval of a harbour extension. For this, says one writer, 'a developing continent must clearly learn to be practical'. In Africa parents have become convinced that education is the key to a good job and family prestige too. But it has its drawbacks. In many countries secondary and college education means that children have to leave the rural areas for the towns and later as mean they are not returning to work on the farms and produce the food. The opportunities and leisure attractions of the towns are too tempting.

Questions

i. From this passage we can tell that colonial education was to
   A. help Africans acquire their independence
   B. help Europeans swiftly rule the Africans
   C. meet the needs of modern African
   D. replace African tradition and culture
   E. Civilise the African children.

ii. The word ‘conform’ in the second paragraph means
   A. comply with rules or general custom
   B. discuss with other members of the society
   C. respect the elders of the society
   D. mix with the rest of members of the society
   E. disagree with the community values.

iii. ‘Elite’ in the fourth paragraph means
   A. a group of ‘selected people

B. a group of colonialists
C. a group of westernised Africans
D. a group of rich people
E. a disadvantaged group.

iv. The traveler in East Africa was surprised (in paragraph three) because
A. he could not expect such young children to prepare such a delicious meal
B. he thought only European children could prepare a meal like that
C. the children could prepare a meal without assistance from an older person
D. the children were not competently supervised when preparing the meal
E. the children were too few to prepare a meal.

v. We can deduce from paragraphs four and five that colonial education was
A. too difficult for the Africans
B. irrelevant to the African situation
C. showing European birds and snow that did not exist in Africa
D. preparing Africans for highly paid jobs
E. preparing Africans for leadership in missions

vi. According to the writer, education in Africa today
A. should train electrical engineers
B. should be practical oriented
C. does not require universities
D. should base on secondary school level only
E. should not prepare a western elite.

vii. The writer says that in traditional education "severe tests of endurance were a more formal part of this education'. The aim of the tests was to train
A. boys and girls not to fear the sight of monsters
B. boys and girls to survive in the cold
C. a Chagga boy not to fear a leopard
D. Bena girls not to run away from the prêt endingly dead women falling at their feet
E. boys and girls to tolerate in extremely difficult conditions.

viii. Nyerere criticised colonial education because
A. it perpetuated the interests of the colonial society
B. it prepared local clerks and junior officials
C. it encouraged university level only
D. it inculcated the values of the colonized society
E. he just hated colonialists.

ix. In the seventh paragraph, the question '... how many universities should a country have, when half of an age group gets no schooling at all?' implies that the writer
A. does not support university education

B. supports university education for children who got no schooling at all
C. encourages more emphasis be put on lower levels of education than the university education
D. discourages lower levels of education for children who got no schooling at all
E. insists that civil servants should just receive good secondary education.

x. From the last paragraph the writer observes that education in many African countries is
A. preparing young people for rural life
B. preparing young people for urban life
C. attracting young people to areas which they are reluctant to leave
D. trying to prepare young people for rural life, but the urban areas are too tempting
E. preparing young people for rural life but they do not like farming.

2. Read and summarize the following passage in two sentences

How can we depend upon gift, loans and investments from foreign countries and foreign companies without endangering our independence? The English people have a saying which says "He who pays the piper calls the tune"

How can we depend upon foreign governments and companies for the major part of our development without giving those governments and countries a great part our freedom and expect them to do as we please? The truth is that we will end up in disappointment.

SECTION B (20 Marks)

PATTERNS AND VOCABULARY

Answer **all** questions in this section

3. Fill in the blanks by writing the correct form of the word from the brackets against the item number.

Tanzania (i)¬¬¬¬¬¬¬¬¬¬¬¬¬_____¬¬¬¬¬¬¬¬¬¬¬¬¬¬¬¬¬ (take) a significant step to (ii)_____ (nation) her means of wealth after Arusha declaration of 1967 and therefore (iii)_____ (embrace) socialism. Tanzania (iv)_____(create) village communities through the (v)_____(village) policy. People (vi)_____ (settle) in such village communities to provide (vii)_____(collect) and joint efforts needed to(viii)_____ (maximum) production of wealth.

4. Choose the correct word from the given list below to fill in the blanks in the following sentences: **beg, pursue, managed, ancestors, display, pangaea, mate, similar, pepper, Gondwanaland**
a) _____is a type of friend one is forced to be with in life

b) _____ has the same meaning as the word portray
c) _____is the first super continent named by scientists
d) _____ are our forefathers and mothers
e) _____means achieved, controlled and or succeeded
f) _____ is the same as the English word request
g) _____ is to undertake
h) _____ is close in meaning with the word resemble

5. In each of these sentences one of the words is wrong. Identify and correct them by writing the correct spelling for each word.
   a) The furniture's are sold at cheaper price.
   b) The equipment is needed at workplace today
   c) We are water the garden now.
   d) I took a loaf of advice from Juma last week
   e) Girls has to liberate on their own from boys' traps
   f) Ali meet me for a discussion last evening.
   g) Muchin formations about a town is gathered through narrations.
   h) Both house girls and house boys knows sufferings.

6. Using the words below, complete the following expressions.

   Purpose, vision, delegates, sex, gender, congregation, choir, legend, ballad, fans, comrades, audience.
   a) A final target that drives your undertaking is known as a _____.
   b) A motive or reason for doing a job or business is named a _____.
   c) A group of people worshiping at church session is collectively called a_____.
   d) A story which may not be true is said to be a _____.
   e) Sent representatives with special mission and interests to negotiate on behalf of a state are called the _____.
   f) A biological trait which distinguishes man from a woman is a _____.
   g) A poem prepared for dramatization on stage is classified as a _____
   h) People belonging to the same political parties or military grade call themselves_____

7. Rewrite the following sentences according to the instructions given after each:
   a) Halima is as tall as Asha. (Split the sentence to make two distinctive sentences)
   b) Juma likes Ugali. Ali likes Ugali too (Join the two sentences by using neither---nor)
   c) " I will come Tomorrow" Musa promised (Report this statement)
   d) The Democratic Republic of Congo invites our country to state celebration

(Passivise the sentence)

SECTION C (30 Marks)

LANGUAGE USE

Answer questions **8, 9** and **two (2)** of the **four (4)** given alternatives in question **10.**

8. Rearrange the following five sentences into a logical sequence to make a meaningful paragraph by writing the corresponding letter in the answer booklet provided. Use the following format for your answers.

Sentence number	1	2	3	4	5
Letter					

A. They were shouting to the bus driver who caused a serious accident.
B. On my way to the market, I met a crowd of people shouting.
C. Finally, the bus driver was taken to the nearby police station.
D. One day, I was asked to go to the market by my mother.
E. Two people who were on the motor bicycle died on the spot.

9. Match each expression in **List A** with the correct word from **List B** by writing its letter beside the item number in the answer booklet(s) provided. Use the following format for your answers.

LIST A	LIST B
i. Plants	A. Community
ii. Pupil	B. A shop
iii. A place in boarding school where students/pupils sleep at night.	C. A dormitory
iv. A place where people buy and sell goods	D. A place where experiments are conducted
	E. A hospital

v. Laboratory .	F. Part of an eye or type of factory G. Is part of an eye or means learner H. A factory or a macro organism

Numbers	I	ii	iii	Iv	v
Letters					

10. Answer **two (2)** questions from **four (4)** alternatives given A, B, C and D.

A. Write a story tilted A DAY SHE WILL NEVER FORGET (At least 250 words)

B. Imagine you are the new appointed Regional Commissioner of Kigoma region. In not less than 200 words write a first speech you would deliver in an official welcoming ceremony at aqua lodge beaches. Focus on your plans as a new president Magufuli' appointee.

C. Write an invitation card to request your sisters' friends to attend on your niece's birthday dinner party. Also write telephone messages from those who will call for excuses. Use fictitious names.

D. Write a letter to the Editor of Guardian Newspaper on the challenges of democracy in Tanzania. Use fictitious name and address

SECTION: D (40 Marks)

RESPONSE TO READING

Answer **two (2)** questions from this section. **One (1)** question should be selected from poetry and the other from novels and short stories or plays.

LIST OFREADINGS

NOVELS AND SHORT STORIES

A Wreath for Fr. Mayer - S.N. Ndunguru (1977), Mkuki na Nyota
Unanswered Cries - Osman Conteh, Macmillan
Passed Like a Shadow - RM Mapalala (2006),' DUP
Spared - S.N. Ndunguru (2004), Mkuki na Nyota
Weep Not Child - Ngugi wa Thiong'o (1987); Heinemann
The Interview - P. Ngugi (2002), .Macmillan

PLAYS

Three Suitors: One. Husband	- O.Mbia (1994),. Eyre Methuen
The Lion and the Jewel	- W. Soyinka (1963),
This Time Tomorrow	- Ngugi wa Thiong'o (1972), Heinemann
The Black Hermit	- Ngugi wa Thiong'o (1968), Heinemann

POETRY

Songs of Lawino and Ocol	- 0.P'Bitek (1979), EAPH
Growing up with Poetry	- D. Rubadiri (ed) (1989), Heinemann
Summons	- R Mahala (1960), TPH

11. How does this poem describe the position and roles of women in our societies? (Give eight points).

Echoes
Me,
I plead guilty
That my woman is a tool
To fulfil my sexual desires,
That she's the source of sexual pleasure.
A garden
Where I plant my deeds
So as to get children
Which are mine.
And not hers,
Who have the right
To inherit my property,
And not her.
I plead guilty
That I have regarded my woman
As my property,
Because I paid the bride price
To her parents.
What did her parents think
When they demanded
The cows
The tank of beer,
The goats,
The money
They knew
That they were making wealth out of me
Because their girl will work for me
Just like a slave.

I plead guilty

That I have inherited the stupidity
Of my slavery age parents,
That I have prolonged inequality.
That I have continued oppression.
Though unknowingly,
I am guilty
For dominating
My woman.
My nation plead guilty
That it has shut an eye
To the rights of women,
That it has kept women
In the kitchen,

In the church,
And at home
To look after children.
My nation pleads guilty
That it has given little room
To the she sex,
That the number of girls at school
Has always been smaller
Than that of boys.

My nation pleads guilty
That it has always given a narrow chance
For women to become
Teachers,
Physicians,
Scientists,
Leaders of the state
And public organizations,
That their percentage
Has always been small.

I agree that
If a woman is to be oppressed
By a man
Who is in turn exploited,
Who is in turn disregarded
By the bourgeoisie,
By the petty bourgeoisie,
Then she's ruined:
She has nothing to live for!

I agree that
To deny them freedom
Means to refuse human equality,
Means to accept capitalism
Deep in our hearts

And also means refusing
The development of the majority

12. Poets compose poems not just for pleasure and entertainment but for addressing human social, political and economic issues. Using Song of Lawino and Ocol present eight points to verify the statement

13. 'Rural life in Africa is itself a separate world, a world with its own life styles and manner'. Using two plays of your choice provide four pieces of evidence from each play to substantiate the said analogy.

14. 'In human society misunderstanding is inevitable and this emanates from the roots of the problem which are evident in many societies'. With reference to two novels discuss the eight major roots of misunderstanding, four roots from each novel.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

013 **GEOGRAPHY**
(For Both School and Private Candidates)

Time: 3Hours **Tuesday, 25th June 2019 p.m.**

Instructions

1. This paper consist of section A, B, C and D with a total of **twelve (12)** questions.
2. Answer **all** questions in section A, B and C and **one (1)** Question from each part of section D.
3. Map extract of Songwe River (sheet 244/3) is provided.
4. Calculators, cellular phone and any unauthorized material are **not** allowed in the examination room.
5. Write your **Examination number** on every page of your answer booklet(s).

SECTION A (25 Marks)

Answer **all** questions in this section.

PHYISCIAL AND MATHEMATICAL GEOGRAPHY

1. For each of the following items i- x choose the correct answer from the given alternative and write its letter beside the item number.
   (i) The East Africa TV channel is conducting an interview at Alaska 75^0W at 7.30 pm Tuesday. At what time and day will the people in Katavi 45^0E be

watching the TV?

A 9.30 p.m Tuesday B 3.30 p.m Wednesday C 11.30 p.m Wednesday
D 12.30 p.m Tuesday E 9.30 am Wednesday

(ii) It occur when the earth passes between the moon and the sun;-
A Solar eclipse B Earth eclipse C Lunar eclipse
D Eclipse of the Earth E Umbra eclipse

(iii) Is the raised part of the ocean floor?

A Mountain B Isenberg C Oceanic ridge
D Oceanic trenches E Deep see plain

(iv) The temperature at Dodoma from mean sea level is 31^0c what will be the temperature of Karatu 2500 meter above sea level.

A 16^0C B 15^0C C 46^0C
D 36^0C E 30.4^0C

(v) An earthquake begins
A Along the Fault or fissure B At the inner core of the earth
C At the epicenter D At the focus
E At the Mantle layer

(vi) The rock formed from the remain of once living organism

A Coral B Marble C Gneiss
D Granite E Humus

(vii) Among of the following are the characteristics of the Tropical climate EXCEPT
A It is found at altitude of 3^0-8^0 N and S
B It compose of two season, dry season and Rain season
C It compose of Tropical forest
D It is found along the western Morgan of the continental
E It compose of season swamps

(viii) The lithosphere of the earth is made up of;-

A Molten materials called magma B Silica and Magnesium

C Iron and nickel (nife) D Iron and zinc

E Silica and Alumina

(ix) It involve chemical reaction when water react with mineral content on rock

A Weathering chemical B Biological weathering

C Chemical weathering D Physical biological weathering

E Physical chemical reaction

(x) Which of the following is correct about the concept of folding

A Formed when two block act against each other on the old sedimentary rock

B Formed when two block act apart from each on the old sedimentary rock

C Formed when two crust act against each other on the young sedimentary rock

D Formed when two crust act against each other on the young igneous rock

E It compose of block that are exposed on the earth surface

2. Match a phrase in list A with the responses in list B by writing the letter of the correct responses beside the item number

LIST A	LIST B
i. The soil	A. The thin layer of the earth know as regolith
ii. Soil PH	B. The top most part of the earth made up with consolidated materials
iii. Soil Texture	C. Soil catena
iv. Soil organic	D. Is the vertical arrangement of individual particles of soil
v. Soil profile	E. is the amount of Alkaline and acidic in the soil measured by PH value scale
	F. When the acidic is high than Alkaline

	G. Derived from the decomposition of plant and animal dead materials H. Compose of Inorganic and Organic matter only I. It depend on the size of the individual particles of soil J. The vertical and Horizontal arrangement of soil

3. (a) With the aid of well labeled diagram explain the formation of the features

   i. Barchans
   ii. Cliff
   iii. Cave
   iv. Bay land

   (b) Explain three necessary conditions for the formation of Barchans

SECTION B: (27 MARKS)

Answer **all** questions in this section

APPLICATION OF STATISTICS, INTRODUCTION TO RESEARCH AND ELEMENTARY SURVEY

4. Study careful the hypothetical data which show the export of goods in Tanzania from 2010 to 2013 in "000"tones

CROPS	2010	2011	2012	2013
COTTON	65	45	70	80
SISAL	35	40	30	35
COFEE	50	70	55	55

a) Name statistical way used to present the data above
b) Present the data above by using a compound bar graph
c) Comment on the export values.

5. (a) (i) What is hypothesis in research
(ii) Explain two types of hypothesis in research

(b) Define the following terms
(i) Sample
(ii) Sampling error

(c) Explain three ways to avoid sampling error.

6. (a) Define the following terms as used in survey

i. Ranging

ii. Chaining

iii. Booking

(b) Why the follower direct the leader left, right and center during survey process (3
points)

SECTION C: (28 Marks)

Answer **all** questions in this section

MAP READING AND PHOTOGRAPH INTERPRETAION

7. Study careful the printed Map extract of Songwe River (Sheet 244/3) then answer the following questions;-
a) Calculate the area covered by Ruanda Town in km2 by using grid square method.
b) Measure the bearing and direction of Muyovinzi to Myavisi station
c) Name three ways' used to show relief on the map
d) Explain three factors for population distribution on the Map
e) Describe the nature of the vegetation shown on the map.
f) Name two (02) natural and two (02) artificial content shown on the map.

8. (a). How a person can determine the desert climatic area on the picture.(four points)

(b) Differentiate aerial oblique from ground oblique photos.
(c) Identify then name type of photograph which:
   i. Does not show horizon
   ii. Taken at slant angle
   iii. Taken at angle less than 90 degree
   iv. Only show front view
   v. Has less scale distortion
   vi. Taken from the air and show top and sides view

SECTION D (20 Marks)

Answer **one (1)** question from each part.

PART I: REGIONAL FOCAL STUDIES

9. Explain six (6) strategies which empowered by Tanzania government in solving problem of Transport and communication sector.

10. "Although Tanzania has many water bodies resources but does not develop in fishing industry" Justify this statement by using six (6) points.

PART II: ENVIRONMENTAL ISSUES, POPULATION AND SETTLEMENTS

11. (a) What is biodiversity ?
    (b) Explain four (4) causes for the loss of biodiversity and propose four (4) measure to solve the problem.

12. Examine six (6) factors which leads to the decreasing of mortality rate in Tanzania.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

012 **HISTORY**
(For Both School and Private Candidates)

Time: 3 Hours **Wednesday,26th June 2019 p.m.**

Instructions

1. This paper consists of sections A, B and C.
2. Answer **all** questions in sections A and B and **three (3)** questions from section C.
3. Cellular phones are **not** allowed in the examination room.
4. All drawings should be in pencil.
5. Write your **Examination Number** on every page of your answer booklet(s).

SECTION A (20 Marks)

Answer **all** questions in this section.

1. For each of the items (i) (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.

   i. There are two levels of looking at history. At the first level history studies man's struggles to obtain fundamental material needs. At the second level history_____
      A. studies past records and human achievements

B. investigates human failures and success in course of production
C. searches for information about past societies in order to discover past cultures
D. studies relationships that develops between man and man, and between man and the environment in course of material production
E. describes the past so that one knows how the present came into being and this gives opportunity to forecast the future.

ii. The large state in Central Africa from 1885 to 1908 in personal union with the Kingdom of Belgium under Leopold II was known as_____
A. Katayamvo
B. Congo Free State
C. Tsonga
D. AmaZulu

iii. Christopher Columba and his crew discovered the present day North America in _____
A. 1652
B. 1498
C. 1492
D. 1698
E. 1693

iv. What was the achievement of man during the late Stone Age?
A. Discovered fire and ate cooked food.
B. Started walking upright using forelimbs.
C. Made and used pebble and chopping tools.
D. Started walking on all four limbs
E. Domesticated animals and wild fruits.

v. Which one among the following places did people extract gold from in West Africa?
A. Bure and Wangara
B. Taghaza and Bilma.
C. Axum and Meroe.
D. Meroe and Egypt
E. Uvinza and Bure.

vi. Which one among the following factors contributed to the rise of states in North Africa?

A. Low level of productive forces.
B. Presence of chartered companies.
C. Hunting and gathering activities.
D. Absence of classes among the people.
E. Trade across Sahara and Islam.

vii. _____is a unifying name for two groups of peoples of Southern Africa who share physical and putative linguistic characteristics distinct from the Bantu majority of the region. Culturally, they are divided into the foraging San, or Bushmen, and the pastoral Khoi, or more specifically Khoikhoi, previously known as Hottentots.
A. Khoisan peoples
B. Herero
C. Namaqua
D. Ovambo,
E. Damara.

viii. The people of African origin in Diaspora supports the development of African development through_____
A. The East African community
B. Organization of African Unity
C. East African Community
D. United Nations Organization
E. Pan Africanism.

ix. Which one of the following best explains the reason which made Tanganyika to use constitution means during her struggle for independence?
A. It was a mandated territory.
B. It was a trusteeship territory.
C. It adopted the Open Door Policy.
D. It was a settler colony.
E. It was under indirect rule policy.

x. The smallest unit of French political administration in French Colonial Africa was headed by a European officer and consisted of several cantons, each of which in turn consisted of several villages, and was instituted in France's African colonies from 1895 to 1946. This subdivision was named as_____
A. Circle system
B. Assimilation

C. Association
D. Direct rule
E. Indirect rule

2. Match the stems in **List A** with the correct responses in **List B** by writing the letter of the correct response beside the item number in the answer booklet provided.

List A	**List B**
i. The Kanem Empire ii. The Soninke people iii. Aksumite Empire iv. Berbers v. Herero Wars vi. Sakalava vii. In 1877 viii. From 11, October, 1899, to May 31, 1902 ix. 1602 x. 1879	A. The name of the kings (or mai, as they called themselves) of the Kanem-Bornu Empire, centered first in Kanem in western Chad, and then, after 1380, in Borno (today's northeastern Nigeria). B. A Mandé people who descend from the Bafour and are closely related to the Imraguen of Mauritania,founders of the ancient empire of Ghana c. 750–1240 CE. Subgroups include the Maraka and Wangara. C. At its height extended across most of present-day Eritrea, northern Ethiopia, western Yemen, southern Saudi Arabia, and Sudan. D. An ethnic group indigenous to North Africa distributed in an area stretching from the Atlantic Ocean to the Siwa Oasis in Egypt, and from the Mediterranean Sea to the Niger River. E. A series of colonial wars between the German Empire and the Herero people of

	German South-West Africa (present-day Namibia, c. 1903–1908). F. An ethnic group of Madagascar that occupies the western edge of the island from Toliara in the south to Sambirano in the north G. The Transvaal Boer republic was forcefully annexed by Britain as part of the attempt to consolidate the states of southern Africa under British rule. H. At its height encompassed an area covering Chad, parts of southern Libya (Fezzan) and eastern Niger, north-eastern Nigeria, and northern Cameroon. I. The Second Boer War started broke out and Great Britain defeated two Boer nations in South Africa: the South African Republic (Republic of Transvaal) and the Orange Free State J. The Dutch East India Company founded K. British settlement at the cape L. The Almoravids invaded Ghana empire M. Mineral revolution in South Africa N. The Boer trek O. Difagine wars

	P. Ngoni migration Q. Cecil Rhodes arrived in Southern Rhodesia R. The Sahel region S. Nama Herero wars

SECTION B (20 Marks)

Answer **all** questions in this section.

3. (a) Draw a sketch map of Africa and locate
   i. The Trans-Atlantic caravan routes
   ii. The cape colony
   iii. The land of Northern Rhodesia
   iv. The city state and country where Fort Jesus was built by the Portuguese
   v. The two African countries which were not colonized.

   (b) Mention five coastal towns in East Africa which became famous because of the commercial contacts between the people of East Africa, Middle and Far East

4. (a) Arrange the following statements in chronological order by writing number 1 to 5 beside the item number.
   i. Afterwards colonialists established colonial economy and colonial relations of production in a colony
   ii. By the beginning of 1900's most colonial powers had successfully suppressed all initial African resistances.
   iii. The Berlin conference of November 1884 to February 1885 paralyzed the scramble for African continent among the imperialist powers
   iv. Then the colonialists established the colonial state to help impose colonial rule among Africans
   v. The rise of monopoly stage of capitalism and imperialism in the mid ninetieth century in Europe made Europe unable to meet economic motives within Europe hence began to fight to secure areas of influence outside Europe particularly in Africa.

(b) The items (i- iv) consist of four sentences which form a complete historical meaning. One of the sentences (A- E) is missing. Identify the missing sentence and write its letter beside the item number.

(i) 1. In 1652, a century and a half after the discovery of the Cape sea route, Jan van Riebeeck established a refreshment station at the Cape of Good Hope at what would become Cape Town, on behalf of the Dutch East India Company.

2. The Dutch transported slaves from Indonesia, Madagascar, and India as labour for the colonists in Cape Town.

3. ¬¬¬

Companies to exploit protectorates

C. Trading companies financed the activities of missionaries in scramble for colonies
D. Missionaries learnt native languages hence became useful interpreters when trading companies were signing bogus treaties with indigenous.
E. William Mackinnon was an explorer who headed Imperial British East African Company.

(iii) 1. West Africans came into contact with European merchants during mercantilism.
2. The early stage of capitalism was the phase of primitive accumulation of capital.
3. _____.
4. During this phase commerce was the most important activity.
A. The motives of European merchants were to stop slave trade.
B. West African societies benefited much from these early contacts.
C. In Europe it was known as the age of discovery and overseas expansion.
D. The red Indians for the first time came to trade in West Africa.
E. In this trade the European slaves were exchanged for African goods.

(iv) 1. Assimilation was introduced in Africa by French
2. _____.
3. The system was similar to German direct rule
3. The system intensified resistance from the natives.
A. Africans used Jumbes and Akidas to represent their problems.
B. Portuguese applied this system in her colonies in Africa.
C. The system improved the living standard of the Africans.
D. The policy banned traditional institutions.
E. The system helped the French to consolidate tribalism.

(v) 1. Colonial transport network ran perpendicular to the coast.
2. Railroad was the chief means of transport to the coast in many colonies
3. Colonialists used the means of transport to facilitate colonial production
4. _____.
A. Colonial transportation concentrated in rural areas.
B. Colonial administrators, labourers and soldiers were

transported to different parts of colony calling for their need and use.

C. There was equal distribution of transport services during colonialism.
D. Transport services led to the higher population growth in colonies.
E. Transportation led to the emergence of civil wars in Africa.

SECTION C (60 Marks)

Answer **three (3)** questions from this section.

5. Discuss the cause of the Boer trek and its consequences to the people of South Africa. Give three points each side.
6. Account for the rise and fall of Mali Empire in Western Sudan. Give three points each side
7. Why did it take too long to end slave trade in East Africa? Give six reasons.
8. With examples elaborate the six weaknesses of colonial education.
9. Decolonization in Portuguese colonies in Africa was tough and complicated. The colonies under Portuguese opted armed struggle instead of peaceful means. Why such colonies waged armed struggle? Give six points
10. Explain six problems facing post-colonial African societies whose rootstrace back to colonial era.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

024 LITERATURE IN ENGLISH

(For Both School and Private Candidates)

Time: 2:30Hours **Friday, 28th June 2019 p.m.**

Instructions

1. This paper consists of sections A, B, C and D with total of twelve (12) questions.
2. Answer **two (2)** questions from section A and **one (1)** question from each of sections B, C and D.
3. Each question carries 20 marks.
4. Cellular phones are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s).

SECTION A (40 Marks)

THEORIES OF LITERATURE

Answer **two (2)** questions from this section.

1. Discuss the four advantages and four disadvantages of written literature over oral literature.

2. Write short notes on the following literary concepts
   a) Elegy
   b) Sonnet
   c) Lyric poem
   d) Tragedy drama
   e) Literary criticism
   f) Briefly explain each of the following concepts as applied in literature
      (i) Resolution
      (ii) Climax
      (iii) Crisis
      (iv) Prologue
      (v) Mystery

3. What is a play? How are plays unique? Give eight areas demonstrating of uniqueness of play and drama.

SECTION B (20 Marks)

PLAYS

Answer **one (1)** question from this section.

TEXTS

The Lion and the Jewel	-	WoleSoyinka
The Trials of Brother Jero	-	WoleSoyinka
The Swamp Dwellers	-	WoleSoyinka
Kinjeketile	-	IbrahimHussein
Dilemma of a Ghost	-	AmaAtaAidoo
The Barbed Wire	-	M.Rugyendo

DedanKimathi - KennethWatene
The Government Inspector- N.Gogol

4. Relate the stories and incidences from the plays you have read with the realities people witness in daily life. Give four points from each play.

5. With references to two plays you have read, show how the conflicts found in the readings have impacted individuals. Give four points from each play.
6. Verify relevance of stage direction in effecting art and content in two plays you have read. Give four points from each play to support your answer.

SECTION C (20 Marks)

NOVELS AND SHORT STORIES

Answer **one (1)** question from this section.

TEXTS:

Tales of AmadouKoumba	-	DavidDiop
Quartet	-	RichardRive
A Walk in the Night and Other Stories	-	AlexLa Guma
Houseboy	-	FerdinandOyono
The Old man and The Medal	-	FerdinandOyono
The Concubine	-	ElechiAmadi
Sundiata: An Epic of Old Mali	-	DjibrilTamsirNiane
Secret Lives	-	NgugiWaThiong'o
The Three Solid Stones	-	MarthaMvungi
Girls at War	-	ChinuaAchebe
Uncle Tom's	-	Children Richard

7. Characters in literary works inspire and disappoint readers. How are your inspired by any two characters from two readings you have done? Give four points for each character.

8. Choose two readings from those read under this section and write down thematic similarities and differences. Give four points from each reading.

9. Discuss eight roots of conflict displayed in two readings of your choice by giving four points from each reading.

SECTION D (20 Marks)

POETRY

Answer **one (1)** question from this section.

TEXT: Selected poems —Tanzania Institute of Education

10. Using two poems demonstrate the eight unique features of poetry as an independent genre of literature.

11. Show how the poet has succeeded to deploy figures of speech to reflect realities in two poems you have studied under this section. Give six points.

12. Discuss the eight messages from any four poems you have read. Give two messages from each poem.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

041 BASIC MATHEMATICS
(For Both School and Private Candidates)

Time: 3Hours Tuesday, 25th June 2019 a.m.

Instructions

1. This paper consists sections A and B.
2. Answer all questions in section A and any **four (4)** questions from section B.
3. All necessary working and answers for each question done must be shown clearly.
4. Mathematical tables and non-programmable calculators may be used.
5. Cellular phones are **not** allowed in the Examination room.
6. Write your **Examination Number** on every page of your answer sheet(s) provided

SECTION A (60 MARKS)

Answer **all** questions in this section.

1. (a) Estimate the value of 4.1×0.083
   (b) Find the fractions of the following
   i. 0.8 3
   ii. 0.125%

2. (a) Solve (i) $(\frac{1}{3})$ x = 243 - 1

   (ii) 288 = 8x2

   (b) If log y +2 log x = 3, express y in term of x.

3. (a) If a * b = a2 + b, find y given that
4* (2 * y) = 25

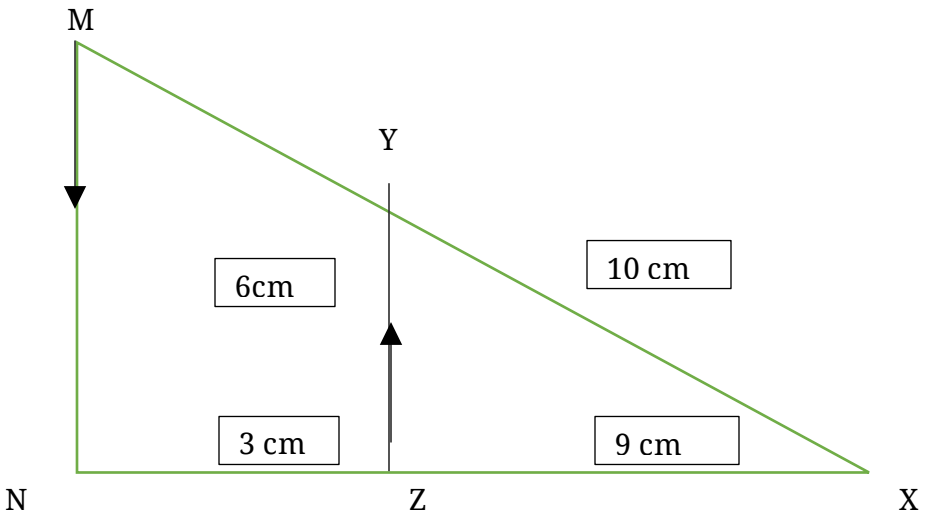
   (b) List all subset of A = {1, 2, 3}.

4. (a) Show that A (-3, 2), B (5, 6) and C (7, 2) are vertices of a right angled

triangle.

(b) Find the unit vector in the direction of vector U = (12, 5)

5. (a) In figure below, calculate
   (i $\overline{MN}$ ii. $\overline{MY}$



(b) Find the radius of a circle which inscribes a regular hexagon with perimeter 50 m.

6. (a) Given that P is inversely proportional
   Q2 When P = 4 and Q = 8, Find the value of Q if P= 16.

(b) A BOOD BUS consumes a diesel at a rate of litre for every 6 km. If the price ofdiesel is sh 2500 per litre. Find the cost of diesel used to travel 960km.

7. (a) X, Y and Z are to share 300,000/= in ratio of 6:4:5 respectively. How much will
   each get.

(b) After reduction of 30% the price of the car become Tsh 490,000. Find its price before reduction.

8. (a) Find the sum of the series
   $-8 - 2 + 4 + 10 + - - - +82.$
   (b) In a G.P the 5th and 7th terms are - 2 and - 18 respectively. Find
   i. Common ratio
   ii. If the common ratio is positive find the sum of the first 10 terms of the series.

9. (a) Without using tables, simplify

$$\frac{\cos 135° + \tan 60° \sin 45°}{\sin 135° \cos 60°}$$

(b) Q and R are two pegs on level ground, and both lie due West of a Flagstaff. The angle of elevation of the top of the flagstaff from P is 45° and from R is 60°.If Pis 24 m from the foot of the flagstaff, Find QR.

10. (a) Solve the equation $49x2 - 25 = 0$

(b)Find the value of x that satisfies the equation $\frac{1350}{X} - \frac{1350}{(X+3)} = 5$

SECTION B (40 MARKS)

Answer **four (4)** questions from this section.

11. At Hijra seminary technical section makes tables and chairs which are processed through the assembly and finish departments. For a particular order from a customer only 48 hours of work available in the assembly and 36 hours in the finishing department. To manufacture one table, it requires 34 hours in assembly and 3 hours in finishing, while a chair requires 4 hours in assembly 2 in finishing. If a chair fetches 40 shillings as profit and table 50 shillings, determine the best combination of tables and chairs to produce so as to get maximum profit.

12. The score of 45 pupils in a physics test were recorded as follows
    30 65 50 62 40 35 64 32 28 59
    60 82 24 35 63 68 46 48 73 92
    28 61 71 36 64 80 61 64 35 76
    73 70 64 61 30.

    a) Construct a frequency distribution table of the given data, taking equal class intervals $21 - 30, 31 - 40, - - -$

    b) Calculate the mean by assumed mean use class boundary $41 - 50$ as a assumed mean.

    c) Draw the cumulative frequency polygon and use it to estimate the median.

13. (a) Find the central angled subtended by an are between the given pair of

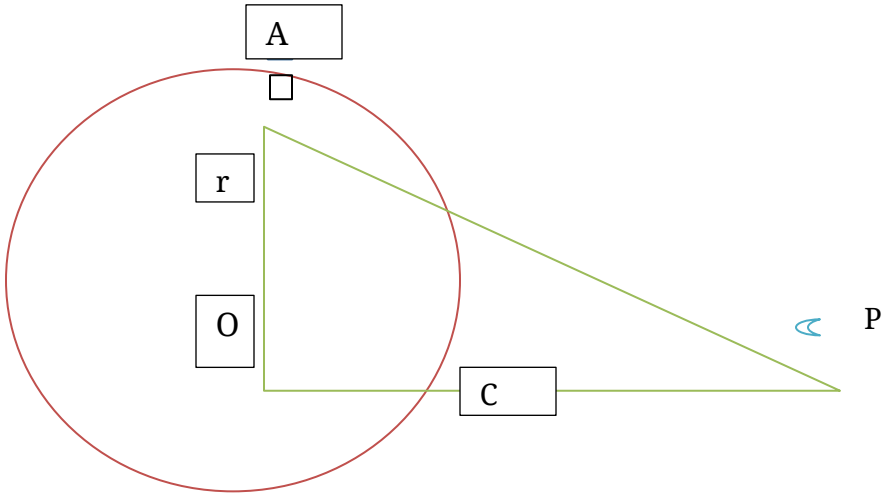
places below.

i. P (18°N , 52°E) and (78°N, 52°E)

ii. X (44°N, 60°E) and(64°N, 60°E)

(b) Find the length of the tangent AP in the figure below; given that O is the centre

ofthe circle r = 5cm and PC = 8 cm.



14. Record the following transactions below in a cash book, post to ledger and then construct a trial balance to check mathematical accuracy of entries.

On 1st February , Ahmed started business with capital in cash of sh 60,000
February 2 purchased goods for cash sh 50,000
February 3 paid cash for transport sh 10,000
February 10 sold goods for cash sh 90,000
February 20 Bought furniture for cash sh 30,000
February 28 paid cash for carriage sh 5000

15. (a) By using inverse method solve the following simultaneous equation

$3x + 2y = 12$

$4x + 5y = 23$

(b) Find the equation of a line $2y = 4x + 10$ after reflection in the line $y = x$

(c) Find the image of the point (-1 ,6) under a rotation of 90° followed by another rotation of 180° anticlock wise.

16. (a) A fraction is written by selecting the numerator from the digits 1, 2, 4, 6 and denominator from 8, 10.

i. Draw a tree diagram to find possibility set of this experiment.

ii. Find the probability that the fraction written is less than $\frac{1}{2}$.

(b) f(x) is defined by:

$$f(x) = \begin{cases} x2 - 5x + 6 \text{ if } 0 \leq x < 4 \\ 6 \text{ if } x \geq 4 \\ -2 \text{ if } x \geq 4 \end{cases}$$

i. Sketch the graph of f(x)

ii. State the domain and range f(x).

iii. Use the graph to find f (5), f (0), f (-1)

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA

ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

031/1 **PHYSICS 2A**

(For Both School and Private Candidates)

Time: 3 Hours **Tuesday, 2nd July 2019 a.m.**

1. You are provided with masses 50g, 100g, 150g, 200g and 250g, spiral spring, stop watch, retort stand.
2.
   a) Suspend the spiral spring to the retort stand.
   b) Load the lower end of the spring with a mass M of 50g and then pull the mass slightly vertically downwards through a short distance from an equilibrium position and release it so that the system executes vertical oscillations of small amplitudes.
   c) Use a stop watch to record the time taken (t) for 20 vertical oscillations and then determine the periodic time (T), hence determine T^2.
   d) Repeat the procedures for the other four masses 100g, 150g, 200g and 250g.
   e) Tabulate your results in the table below

Mass (g)	Tame taken t(s)	Time period T (s)	T^2 (s^2)
50			
100			
150			
200			
250			

   f) Draw a diagram for this experiment
   g) It is found that the period T of oscillations and the spring constant K are related by the equation $T^2 = 4\pi^2\frac{(m + s)}{K}$ where S is the effective mass of the spring
      i. Plot a graph of T^2 against M

ii. Calculate the slope of the graph

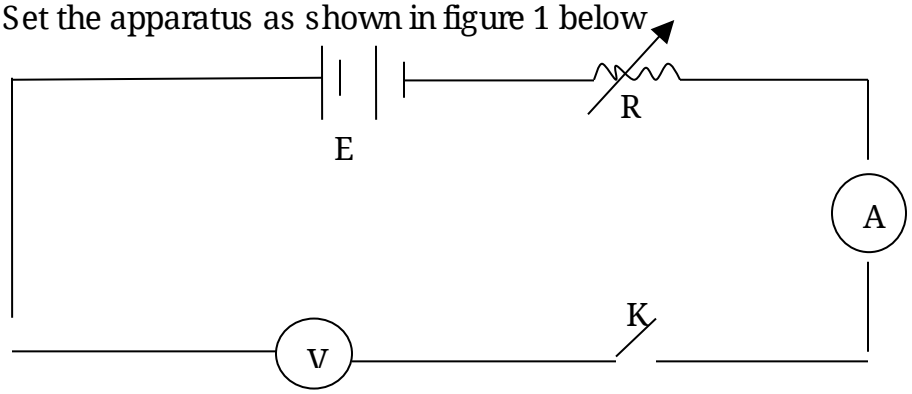
iii. Using the given equation and the graph, determine the value of K and S.

h) (i) Mention one source of error and its precaution.

(ii) Suggest the aim of this experiment.

3. You are provided with battery, Resistance box, Ammeter, Voltimeter, key and connecting wires proceed as follow:

a) Set the apparatus as shown in figure 1 below



b) (i) Close the switch and put R = 1Ω , quickly read and record the value of V and I

(ii) Repeat the procedures by R = 3 Ω , 5 Ω , 7 Ω , and 9 Ω ,

(iii) Tabulate your results.

c) (i) Plot a graph of V against I

(ii) Determine the slope of your graph

(iii) What is the physical meaning of the slope obtained in c (ii) above?

d) If experiment obey's equation V = E Ir, where each symbol carries usual meaning, determine the value of E and r

e) What is the aim of this experiment

f) State two (2) sources of errors and how to minimize it.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

031/1 **PHYSICS 1**
(For Both School and Private Candidates)

Time: 3 Hours **Saturday, 29th June 2019 a.m.**

Instructions

1. This paper consists of section A,B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in section A and B and only **one (1)** question in section C.
3. Calculators, cellular phones are not allowed in the examination.
4. Write your **Examination Number** on every page of your answer booklet(s).
5. The following constants may be used.
    - Density of frash water = $1g/cm^3$
    - Relative density of sea water = 1.03
    - Acceleration due to gravity = $10m/s^2$

SECTION A (30 Marks)

Answer **all** questions in this section.

1. Choose the must correct answer for (i-x) and write your answer in the booklet provided

   i. A roung measure of the size of a molecule can be obtained by
      A. Measuring the height to which water rises in a narrow capillary tube
      B. Finding the speed with which bromine vapor spreads in air
      C. Observing Brownian motion of smoke particles
      D. Measuring the area of the circle into which a small oil drop spreads on water.
      E. Measuring the friction force between the molecules of a solid

   ii. Colors are produced when white light passes through a glass prism because
      A. The light wave interfere
      B. The glass colors the light
      C. Different colors travel at different speed in glass
      D. The different colors are filtered
      E. Diffraction of light occurs

   iii. A rectangular solid box has dimensions 2m by 1m by 0.5m and weighs 100N . when the box displaced on flat ground the maximum pressure exerted is
      A. $100.54N/m^2$
      B. $200N/M^2$
      C. $100N/M^2$
      D. $50N/M^2$
      E. $1000N/M^2$

   iv. When a gas is compressed at constant temperature, the gas molecule
      A. Move faster so that the pressure is increased
      B. Move at the same speed so that the pressure unchanged

C. Gain kinetic energy
D. Increase slighting
E. Move faster so that the pressure is lowered

v. When total internal reflection is just about to occur at an air – water boundary as the incident angle is increased from zero, the refracted ray then
A. Travels in the water at 90 to the normal
B. Travels in air along the normal
C. Makes the critical angle with the normal in
D. Travels in the air at 90 to the normal
E. Makes an angle of 90 with the incident ray

vi. A horizontal electron beam passes between two parallel horizontal plates X and Y with X above Y. when a high tension battery is connected with its negative pole to x and its positive pole Y the beam is deflected
A. Upwards
B. Side ways
C. Down wards
D. Up wards at 45 to the horizontal
E. Down wards at 45 horizontal

vii. An airship is floating stationary high above the grounds. In this case
A. The air up thrust is to the airship weight
B. The air temperature inside the airship is equal to the air temperature of outside
C. The air density outside the airship is greater than the air density inside
D. The air density outside the airship is less than the air density outside
E. The air up thrust is greater than the airship weight

viii. A vibrating string has a tension of 40N and produces a note of 200Hz when plucked in the middle. When the length of string is unaltered and the tension is increased to 160N the frequency
A. 1600HZ B. 800HZ C. 400HZ
D. 200HZ E. 25HZ

ix. A wire X is half the diameter and half the length of wire Y of similar material. The resistance od X to that of Y is
A. 8:1 B. 4:1 C. 2:1 D. 1:1 E. 1;4

x. When a beam of alpha particles was fired at a thin gold film, some of the particles were considerably deflected. This experiment shows that
A. Alpha – particles are very light
B. A gold nucleus contains alpha – particles
C. Electrons in a gold nucleus attract alpha particles
D. A gold nucleus is concentrated in a small it
E. A gold nucleus emits alpha – particles

2. Match the items from **LIST A** that correspond with item from **LIST B** and write the answer in the booklet provided.

LIST A	LIST B
i. Simple microscope ii. Optical instrument iii. Angular magnification iv. Short sightednedd v. Accommodation vi. Hyper myopia vii. Diaphrasm viii. Shutter ix. Lens x. Projection lantern	A. Mechanism of viewing long and short distance objects B. Mechanical device that acts as a gate C. Referred to as a magnifying glass D. Ration of image distance to object distance E. $M = \beta/\propto$ F. Is a light sensitive surface of the camera G. Occurs when the eyeball is longer than normal along the horizontal axis from the lens of the retina H. This deface arises from a shortening of the eyeball I. Used to display a large image on a screen J. Determines the amount of light that passes through the lens by changing the size of the of entire K. Image – forming device on a camera L. Computers and television M. Devices which are used to view small or distant objects more clearly

3. For each the items (i-x) fill in the back spaces by writing the correct answer on the booklet provided
   i. _____ is an instrument used for measuring the relative humidity of the air
   ii. Sound waves in air produce _____ and rarefactions
   iii. Connecting wire is made of pure copper which has a _____ resistance
   iv. _____ is produced at right angles to the plane of a current – carrying conductor
   v. The velocity ration is equal to number of _____
   vi. _____ is a measure of inertia and _____ produces acceleration
   vii. The quantity of heat required to raise that temperature of unit mass of a

substance is called _____

viii. The ratio of $\frac{\sin i}{\sin r}$ is termed as _____

ix. _____ is the splitting of a heavy atom such as uranium, into two heavy parts

x. Device which is used in battery chargers to provide direct current (d.c) from the (a.c) mains supply is _____

SECTION B (60 Marks)

Answer **all** questions in this section.

4. (a)(i) What is meant by the term "moment of a force" give its SI Unit.

(ii) How can the moment of force be increased considerably in practical life? Give two examples.

(b)(i) List two conditions for a body to the in equilibrium.

(ii) Briefly explain why the handle of a door is near its outside edge?

(c)(i) What is meant by centre of gravity?

(ii) It is found that a uniform wooden lath 100cm ling mass of 95g can be balanced on a knife edge when a 5g mass is hung 10cm from one end. How for is the knife – edge from the centre of the lath

5. (a) (i) Define principal focus and focal length

(ii) Explain how mirage occurs

(iii) Why does the swimming pool appear much shallower than its actual depth?

(b) (i) Explain the meaning of the term dispersion of white light"

(ii) Explain the terms "Primary color" and complementary color as applied to colors of light

(c) (i) Show that to obtain an image with a magnification of M using a concave mirror with a focal length f, the object distance 'u' is given by

$$u = \left(\frac{M+1}{M}\right)f$$

(ii) The image in a converging lens is upright and magnified four times. Calculate the object distance if the focal length is 20cm

6. (a) (i) Explain the terms interference and diffraction of waves

(ii) Explain why strings of different thickness are used on a stringed instrument such as violin?

(b)(i) Differentiate between echo and amplitude

(ii) What is meant by sonometer?

(c) (i) Explain why notes of the some pitch played on a violin and a flute sound different.

(ii) A pipe closed at one end has a length of 100mm. if the velocity of sound in the air of the pipe is 340m/s, calculate (a) the fundamental and (b) The first overtone

(d) Give one reason why the frequency of the fundamental may alter during the day

7. (a) (i) Define the term momentum

(ii) Briefly explain why seat – salts are designed to stretch in a collision

(b) (i) What is meant by incretion

(ii) Sand falls gently at a constant rate of 50g/s on to a horizontal belt moving steadily 40cm find the force in Newton's on the salt explode and sand.

(c) A stationary bomb of mass 5000g explodes into one part A of mass 2kg flying off with a velocity of 60m/s and another part B of mass 3kg fling off with a velocity 'v' in the opposite direction calculate

(i) V (ii) The total kinetic energy produced by the engision

8. (a) (i) Define the term nuclear fusion and give one example

(ii) In 168 seconds the activity of thoron falls to one – eighth of its original value. What its half-life explain this term

(b) (i) Using a sketch, explain the effects of electric field on alpha par trice, date particle and garama rays

(ii) A radioactive nucleons is demoted by the symbol $^{288}_{92}M$. Write down how you would expect the nucleus to be composed and state the composition of the nucleons at the end of each of the following stages of disintegration

1. The emission of alpha particle
2. The further emission of Beta particle
3. The further emission of gamma radiation

(c) Explain how nuclear radiation affects the following

i. Gastrointestinal tract
ii. Reproductive organs
iii. Eyes

9. (a) Explain briefly how the following are produce

i. A p- type semiconductor
ii. An n- type semiconductor
iii. Transistors

(b) Discuss the mode of action of the following
i. A p-n junction
ii. Diodes in half – wave rectification of a.c current

(c) Define the following terms
i. Conductor
ii. Semiconductor
iii. Insulator

(d) Describe the effects of temperature on conductors, semiconductors and insulation the energy band concept.

SECTION C (10 Marks)

Answer **one (1)** question from this section.

10. (a) Define the following term
(i) Star
(ii) Planet
(iii) Constellation

(b) (i) What is the importance of installation?
(ii) Explain how ocerntides occur

(c) (i) What is an earth quake?
(ii) Give three precautions that can be taken against earth quake hazards
(iii) Describe two effects of volcanoes

(d) Explain the meaning of the following term as used on earthquakes
(i) Hypocentre (ii) Epicentre

11. (a) (i) State law of floatation
(ii) Explain why ship float but a piece of iron sink

(b) (i) State two conditions for a body to float
(ii) An ordinary hydrometer of mass 20g floats with 3cm of its stem out of water. The area of cross – section of the stem is $0.75cm^2$. Find the total volume of the hydrometer and length of the stem above the surface when it floats in a liquid of relative density 1.4.

(c)(i) Define up thrust
(ii) A ship of mass 1200tones floats in sea – water. What volume of sea-water does it displace? If the ship enters fresh water what mass of cargo must be unconded so that the some volume of water is displaced as

before?

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT
FORM FOUR ZAIM MOCK EXAMINATION

131/2A PHYSICS 2A

(ACTUAL PRACTICAL)

ONE MONTHY ADVANCE INSTRUCTIONS

1.0 IMPORTANT:

I. **GREAT CARE MUST BE TAKEN NOT TO DIVULGE THESE INSTRUCTIONS TO BOTH CANDIDATES AND UNAUTHORIZED PERSONS EITHER DIRECTLY OR INDIRECTLY.**

II. NOTE THAT YOU WILL NOT BE ALLOWED TO OPEN THE ENVELOPE CONTAINING THE QUESTION PAPERS BEFORE THE TIME OF COMMENCEMENT OF THE EXAMINATION.

III. THERE WILL BE ON 24 HRS ADVANCE INSTRUCTIONS FOR THIS PAPER.

(Some of them will be used in the examination for physics practical)

LIST OF APPARATUS

1. STOP WATCH
2. METER BRIDGE WITH ITS ACCESSORIES
3. OFFICE PINS
4. PENDULUM BOB
5. GLASS BLOCK OF DIMENSIONS 10.2CM X 6.4CM X 0.2CM
6. CLAMP
7. RETORT STAND
8. AMMETER
9. METRE RULE
10. VOLTIMETER
11. CONNECTING WIRE
12. CONSTANTAN WIRE SWG-28
13. NICHROME WIRE SWG-28
14. THUMB PINS
15. SPIRAL SPRINGS
16. KEY
17. 2 – DRY CELL SIZE D OF E.M.F 1.5V (PANASONIC)
18. COTTON THREAD
19. SLOTTED MASSES 20G, 40G, UP TO 100G.
20. RESISTANCE BOX (1Ω-10Ω)
21. RHEOSTAT (VARIABLE RESISTOR)
22. SCALE PAN
23. G- CLAMP
24. CORK

25. 5- WHITE PAPERS
26. UNKNOWN MASS LABELLED M
27. PLANE MIRROR
28. DRAWING BOARD
29. PROTRACTOR
30. TRANSPARENT RULER OF 30CM
31. A KNIFE EDGE
32. UNKNOWN RESISTANCE OF 4Ω LABELLED Rx
33. POTENTIOMETER
34. 1- DRY CELL SIZE AA
35. BEAKER (250MLS)
36. EUREKA CAN
37. MEASURING CYLINDER $10CM^3$
38. GLASS PRISM

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT
FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

131/2A PHYSICS 2A

24 – HOURS INSTRUCTIONS

(For Both School and Private Candidates)

In addition to normal fittings of physics laboratory.

PREPARE THE FOLLOWING APPARATUS / INSTRUMENTS.

1. (i) Retort stand with its accessories.
   (ii) Masses 50g, 100g, 150g, 200g and 250g.

(iii) Spiral spring.

(iv) Stop watch.

2. (i) Ammeter.

(ii) Voltimeter.

(iii) Connecting wires.

(iv) Key.

(v) Resistance box range (1n - 10n).

(vi) 2 - Dry cells size D (National panasonic) E = 1.5V.

(vii) Battery holder.